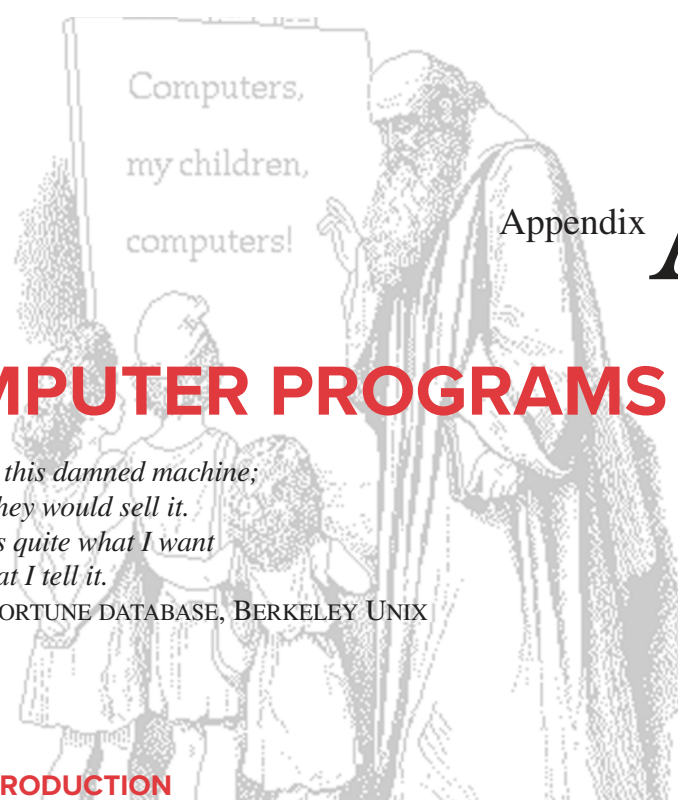




Computers,
my children,
computers!

Appendix

A

COMPUTER PROGRAMS

*I really hate this damned machine;
I wish that they would sell it.
It never does quite what I want
But only what I tell it.*

FROM THE FORTUNE DATABASE, BERKELEY UNIX

A.0 INTRODUCTION

In addition to the downloadable version of the commercial simulation program Working Model, there are three computer programs, written by the author, downloadable with this text: programs LINKAGES, MATRIX, and DYNACAM. These are student editions of the professional versions of these programs for educational use only. For commercial applications, professional versions with extended capabilities are available at <http://www.designofmachinery.com>. Program LINKAGES is based on the mathematics derived in Chapters 4 to 7 and 10 to 14 and use the equations presented therein to solve for position, velocity, acceleration, forces, and torques in fourbar, fivebar, sixbar, and slider linkages and IC engines. Program DYNACAM is a cam design program based on the mathematics derived in Chapters 8 and 15. Program MATRIX is a general linear simultaneous equation solver. All have similar choices for the display of output data in the form of tables and plots. All the programs are designed to be user friendly and reasonably “crashproof.” The author encourages users to email reports of any “bugs” or problems encountered in their use to him at norton@wpi.edu.

To obtain these programs and the other videos and files provided with the book, you need to register as a **student using the book** on the website shown above. Note that I personally review all applications for access to this protected site, and if a student does not fill out the application completely and correctly according to the instructions, then they will be denied access.

Learning Tools

All the custom programs provided with this text are designed to be learning tools to aid in the understanding of the relevant subject matter and *are specifically not intended to be used for commercial purposes in the design of hardware and must not be so used*. It is

quite possible to obtain inappropriate (but mathematically correct) results to any problem solved with these programs, due to incorrect or inappropriate input of data. The user is expected to understand the kinematic and dynamic theory underlying the program's structure and to also understand the mathematics on which the program's algorithms are based. This information on the underlying theory and mathematics is derived and described in the noted chapters of this text. Most equations used in the programs are derived or presented in this textbook.

Disclaimer and Limitations on Use

Student editions of these programs are made available with this book and carry a limited-term license restricted to educational use in course work for up to 2 years. If you wish to use the program for the benefit of a company or for any commercial purpose, then you must obtain the professional edition of the same program. **The student editions may not be used commercially!** The professional editions typically offer more features and better accuracy than the student editions. Commercial software for use in design or analysis needs to have built-in safeguards against the possibility of the user providing incorrect, inappropriate, or ridiculous values for input variables, in order to guard against erroneous results due to user ignorance or inexperience. **The student editions of the custom programs provided with this text are not commercial software and deliberately do not contain such safeguards against improper input data**, on the premise that to do so would “short-circuit” the student’s learning process. We learn most from our failures. These programs provide an educational environment to explore failure of your designs “on paper” and in the process to come to a more thorough and complete understanding of the subject matter. **The author and publisher are not responsible for any damages which may result from the use or misuse of these programs.**

A.1 GENERAL INFORMATION

Hardware/System Requirements

These programs will run in Windows 2000/NT/XP/Vista/Windows 7/8/10, but settings changes are needed in Vista as described in the installation instructions. All programs will operate properly in both 32- and 64-bit operating systems. In Windows 10, you may have to run them as Windows 7 applications.

Installing the Software

The install.exe files contain the executable program files plus all necessary Dynamic Link Library (DLL) and other ancillary files needed to run the programs. Run the Install file for each program to install all of its files on your hard drive. The program name will appear in the list under the *Start/Program/Design of Machinery* menu after installation and can be run from there. DYNACAM and LINKAGES can be updated from that menu also. Use the Check for Updates link within the program’s folder in the Start menu

User Manual

User manuals are accessible from the programs’ help menus. Tutorial videos are also in some programs. Instructional videos are also accessible from the help menus within the programs when the computer is connected to the Internet.

Appendix **B**

MATERIAL PROPERTIES

These tables are for selected engineering materials. Many other alloys are available.

The following tables contain approximate values for strengths and other specifications of a variety of engineering materials compiled from various sources. In some cases, the data are minimum recommended values, and in other cases data are from a single test specimen. These data are suitable for use in the engineering exercises contained in this text but should not be considered as statistically valid representations of specifications for any particular alloy or material. The designer should consult the materials' manufacturers for more accurate and up-to-date strength information on materials used in engineering applications or conduct independent tests of the selected materials to determine their ultimate suitability to any application.

Table No. Description	
B-1	Physical Properties of Some Engineering Materials
B-2	Mechanical Properties of Some Wrought-Aluminum Alloys
B-3	Mechanical Properties of Some Carbon Steels
B-4	Mechanical Properties of Some Cast-Iron Alloys
B-5	Properties of Some Engineering Plastics

TABLE B-1 Physical Properties of Some Engineering Materials

Data from Various Sources.* These Properties are Essentially Similar for All Alloys of the Particular Material

Material	Modulus of Elasticity <i>E</i>		Modulus of Rigidity <i>G</i>		Poisson's Ratio ν	Weight Density γ	Mass Density ρ	Specific Gravity
	Mpsi	GPa	Mpsi	GPa		lb/in ³	Mg/m ³	
Aluminum alloys	10.4	71.7	3.9	26.8	0.34	0.10	2.8	2.8
Beryllium copper	18.5	127.6	7.2	49.4	0.29	0.30	8.3	8.3
Brass, bronze	16.0	110.3	6.0	41.5	0.33	0.31	8.6	8.6
Copper	17.5	120.7	6.5	44.7	0.35	0.32	8.9	8.9
Iron, cast, gray	15.0	103.4	5.9	40.4	0.28	0.26	7.2	7.2
Iron, cast, ductile	24.5	168.9	9.4	65.0	0.30	0.25	6.9	6.9
Iron, cast, malleable	25.0	172.4	9.6	66.3	0.30	0.26	7.3	7.3
Magnesium alloys	6.5	44.8	2.4	16.8	0.33	0.07	1.8	1.8
Nickel alloys	30.0	206.8	11.5	79.6	0.30	0.30	8.3	8.3
Steel, carbon	30.0	206.8	11.7	80.8	0.28	0.28	7.8	7.8
Steel, alloys	30.0	206.8	11.7	80.8	0.28	0.28	7.8	7.8
Steel, stainless	27.5	189.6	10.7	74.1	0.28	0.28	7.8	7.8
Titanium alloys	16.5	113.8	6.2	42.4	0.34	0.16	4.4	4.4
Zinc alloys	12.0	82.7	4.5	31.1	0.33	0.24	6.6	6.6

* Properties of Some Metals and Alloys, International Nickel Co., Inc., NY; *Metals Handbook*, American Society for Metals, Materials Park, OH.**TABLE B-2 Mechanical Properties of Some Wrought-Aluminum Alloys**

Data from Various Sources.* Approximate Values. Consult Manufacturers for More Accurate Information

Wrought-Aluminum Alloy	Condition	Tensile Yield Strength (2% Offset)		Ultimate Tensile Strength		Fatigue Strength at 5E8 Cycles		Elongation over 2 in	Brinell Hardness
		kpsi	MPa	kpsi	MPa	kpsi	MPa	%	HB
1100	Sheet annealed	5	34	13	90			35	23
	Cold rolled	22	152	24	165			5	44
2024	Sheet annealed	11	76	26	179			20	—
	Heat treated	42	290	64	441	20	138	19	—
3003	Sheet annealed	6	41	16	110			30	28
	Cold rolled	27	186	29	200			4	55
5052	Sheet annealed	13	90	28	193			25	47
	Cold rolled	37	255	42	290			7	77
6061	Sheet annealed	8	55	18	124			25	30
	Heat treated	40	276	45	310	14	97	12	95
7075	Bar annealed	15	103	33	228			16	60
	Heat treated	73	503	83	572	14	97	11	150

* Properties of Some Metals and Alloys, International Nickel Co., Inc., NY; *Metals Handbook*, American Society for Metals, Materials Park, OH.

TABLE B-3 Mechanical Properties of Some Carbon Steels

Data from Various Sources.* Approximate Values. Consult Manufacturers for More Accurate Information

SAE / AISI Number	Condition	Tensile Yield Strength (2% Of fset)		Ultimate Tensile Strength		Elongation over 2 in	Brinell Hardness
		kpsi	MPa	kpsi	MPa	%	HB
1010	Hot rolled	26	179	47	324	28	95
	Cold rolled	44	303	53	365	20	105
1020	Hot rolled	30	207	55	379	25	111
	Cold rolled	57	393	68	469	15	131
1030	Hot rolled	38	259	68	469	20	137
	Normalized @ 1650°F	50	345	75	517	32	149
	Cold rolled	64	441	76	524	12	149
	Q&T @ 1000°F	75	517	97	669	28	255
	Q&T @ 800°F	84	579	106	731	23	302
	Q&T @ 400°F	94	648	123	848	17	495
1035	Hot rolled	40	276	72	496	18	143
	Cold rolled	67	462	80	552	12	163
1040	Hot rolled	42	290	76	524	18	149
	Normalized @ 1650°F	54	372	86	593	28	170
	Cold rolled	71	490	85	586	12	170
	Q&T @ 1200°F	63	434	92	634	29	192
	Q&T @ 800°F	80	552	110	758	21	241
	Q&T @ 400°F	86	593	113	779	19	262
1045	Hot rolled	45	310	82	565	16	163
	Cold rolled	77	531	91	627	12	179
1050	Hot rolled	50	345	90	621	15	179
	Normalized @ 1650 °F	62	427	108	745	20	217
	Cold rolled	84	579	100	689	10	197
	Q&T @ 1200°F	78	538	104	717	28	235
	Q&T @ 800°F	115	793	158	1089	13	444
	Q&T @ 400°F	117	807	163	1124	9	514
1060	Hot rolled	54	372	98	676	12	200
	Normalized @ 1650°F	61	421	112	772	18	229
	Q&T @ 1200°F	76	524	116	800	23	229
	Q&T @ 1000°F	97	669	140	965	17	277
	Q&T @ 800°F	111	765	156	1076	14	311
1095	Hot rolled	66	455	120	827	10	248
	Normalized @ 1650 °F	72	496	147	1014	9	13
	Q&T @ 1200°F	80	552	130	896	21	269
	Q&T @ 800°F	112	772	176	1213	12	363
	Q&T @ 600°F	118	814	183	1262	10	375

* SAE Handbook, Society of Automotive Engineers, Warrendale, PA; Metals Handbook, American Society for Metals, Materials Park, OH.

TABLE B-4 Mechanical Properties of Some Cast-Iron Alloys

Data from Various Sources.* Approximate Values. Consult Manufacturers for More Accurate Information

Cast-Iron Alloy	Condition	Tensile Yield Strength (2% Offset)		Ultimate Tensile Strength		Compressive Strength		Brinell Hardness
		kpsi	MPa	kpsi	MPa	kpsi	MPa	HB
Gray cast iron—Class 20	As cast	—	—	22	152	83	572	156
Gray cast iron—Class 30	As cast	—	—	32	221	109	752	210
Gray cast iron—Class 40	As cast	—	—	42	290	140	965	235
Gray cast iron—Class 50	As cast	—	—	52	359	164	1131	262
Gray cast iron—Class 60	As cast	—	—	62	427	187	1289	302
Ductile iron 60-40-18	Annealed	47	324	65	448	52	359	160
Ductile iron 65-45-12	Annealed	48	331	67	462	53	365	174
Ductile iron 80-55-06	Annealed	53	365	82	565	56	386	228
Ductile iron 120-90-02	Q & T	120	827	140	965	134	924	325

* *Properties of Some Metals and Alloys*, International Nickel Co., Inc., NY; *Metals Handbook*, American Society for Metals, Materials Park, OH.**TABLE B-5 Properties of Some Engineering Plastics**

Data from Various Sources.* Approximate Values. Consult Manufacturers for More Accurate Information

Material	Approximate Modulus of Elasticity $E^{\dagger}$		Ultimate Tensile Strength		Ultimate Compressive Strength		Elongation over 2 in	Max Temp	Specific Gravity
	Mpsi	GPa	kpsi	MPa	kpsi	MPa	%	°F	
ABS	0.3	2.1	6.0	41.4	10.0	68.9	5–25	160–200	1.05
20–40% glass filled	0.6	4.1	10.0	68.9	12.0	82.7	3	200–230	1.30
Acetal	0.5	3.4	8.8	60.7	18.0	124.1	60	220	1.41
20–30% glass filled	1.0	6.9	10.0	68.9	18.0	124.1	7	185–220	1.56
Acrylic	0.4	2.8	10.0	68.9	15.0	103.4	5	140–190	1.18
Fluoroplastic (PTFE)	0.2	1.4	5.0	34.5	6.0	41.4	100	330–350	2.10
Nylon 6/6	0.2	1.4	10.0	68.9	10.0	68.9	60	180–300	1.14
Nylon 11	0.2	1.3	8.0	55.2	8.0	55.2	300	180–300	1.04
20–30% glass filled	0.4	2.5	12.8	88.3	12.8	88.3	4	250–340	1.26
Polycarbonate	0.4	2.4	9.0	62.1	12.0	82.7	100	250	1.20
10–40% glass filled	1.0	6.9	17.0	117.2	17.0	117.2	2	275	1.35
HMW polyethylene	0.1	0.7	2.5	17.2	—	—	525	—	0.94
Polyphenylene oxide	0.4	2.4	9.6	66.2	16.4	113.1	20	212	1.06
20–30% glass filled	1.1	7.8	15.5	106.9	17.5	120.7	5	260	1.23
Polypropylene	0.2	1.4	5.0	34.5	7.0	48.3	500	250–320	0.90
20–30% glass filled	0.7	4.8	7.5	51.7	6.2	42.7	2	300–320	1.10
Impact polystyrene	0.3	2.1	4.0	27.6	6.0	41.4	2–80	140–175	1.07
20–30% glass filled	0.1	0.7	12.0	82.7	16.0	110.3	1	180–200	1.25
Polysulfone	0.4	2.5	10.2	70.3	13.9	95.8	50	300–345	1.24

* *Modern Plastics Encyclopedia*, McGraw-Hill, New York; *Machine Design Materials Reference Issue*, Penton Publishing, Cleveland, OH.[†] Most plastics do not obey Hooke's law. These apparent moduli of elasticity vary with time and temperature.

Appendix C

GEOMETRIC PROPERTIES

**DIAGRAMS AND FORMULAS TO CALCULATE THE FOLLOWING
PARAMETERS FOR SEVERAL COMMON GEOMETRIC SOLIDS**

V = volume

m = mass

C_g = location of center of mass

I_x = second moment of mass about x axis = $\int (y^2 + z^2) dm$

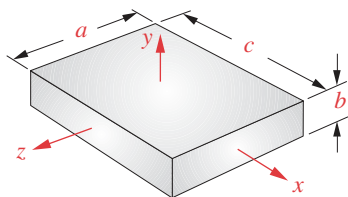
I_y = second moment of mass about y axis = $\int (x^2 + z^2) dm$

I_z = second moment of mass about z axis = $\int (x^2 + y^2) dm$

k_x = radius of gyration about x axis

k_y = radius of gyration about y axis

k_z = radius of gyration about z axis



(a) Rectangular prism

$$V = abc$$

$$x_{CG} @ \frac{c}{2}$$

$$I_x = \frac{m(a^2 + b^2)}{12}$$

$$k_x = \sqrt{\frac{I_x}{m}}$$

$$m = V \cdot \text{mass density}$$

$$y_{CG} @ \frac{b}{2}$$

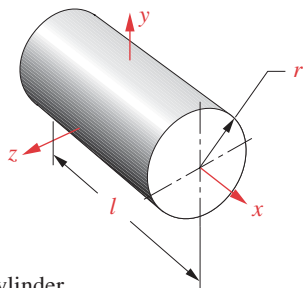
$$I_y = \frac{m(a^2 + c^2)}{12}$$

$$k_y = \sqrt{\frac{I_y}{m}}$$

$$z_{CG} @ \frac{a}{2}$$

$$I_z = \frac{m(b^2 + c^2)}{12}$$

$$k_z = \sqrt{\frac{I_z}{m}}$$



(b) Cylinder

$$V = \pi r^2 l$$

$$x_{CG} @ \frac{l}{2}$$

$$I_x = \frac{mr^2}{2}$$

$$k_x = \sqrt{\frac{I_x}{m}}$$

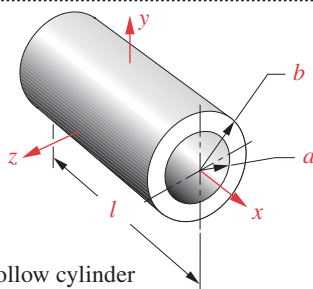
$$m = V \cdot \text{mass density}$$

$$y_{CG} \text{ on axis}$$

$$z_{CG} \text{ on axis}$$

$$I_y = I_z = \frac{m(3r^2 + l^2)}{12}$$

$$k_y = k_z = \sqrt{\frac{I_y}{m}}$$



(c) Hollow cylinder

$$V = \pi(b^2 - a^2)l$$

$$x_{CG} @ \frac{l}{2}$$

$$I_x = \frac{m(a^2 + b^2)}{2}$$

$$k_x = \sqrt{\frac{I_x}{m}}$$

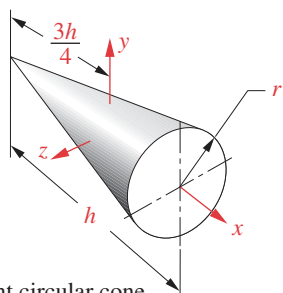
$$m = V \cdot \text{mass density}$$

$$y_{CG} \text{ on axis}$$

$$z_{CG} \text{ on axis}$$

$$I_y = I_z = \frac{m(3a^2 + 3b^2 + l^2)}{12}$$

$$k_y = k_z = \sqrt{\frac{I_y}{m}}$$



(d) Right circular cone

$$V = \pi \frac{r^2 h}{3}$$

$$x_{CG} @ \frac{3h}{4}$$

$$I_x = \frac{3}{10} mr^2$$

$$k_x = \sqrt{\frac{I_x}{m}}$$

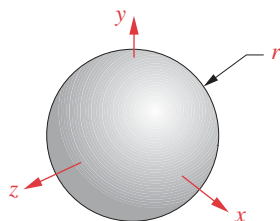
$$m = V \cdot \text{mass density}$$

$$y_{CG} \text{ on axis}$$

$$z_{CG} \text{ on axis}$$

$$I_y = I_z = \frac{m(12r^2 + 3h^2)}{80}$$

$$k_y = k_z = \sqrt{\frac{I_y}{m}}$$



(e) Sphere

$$V = \frac{4}{3} \pi r^3$$

$$x_{CG} \text{ at center}$$

$$I_x = I_y = I_z = \frac{2}{5} mr^2$$

$$k_x = k_y = k_z = \sqrt{\frac{I_x}{m}}$$

$$m = V \cdot \text{mass density}$$

$$y_{CG} \text{ at center}$$

$$z_{CG} \text{ at center}$$

Appendix D

SPRING DATA

The following catalog pages of helical compression and extension spring data were provided courtesy of the *Hardware Products Co., Chelsea, Massachusetts*

<http://www.hardwareproducts.com/>

Other spring information can be found on the Web at:

<http://www.leespring.com/>

<http://www.cookspring.com/>

<http://www.allrite.com/>

<http://www.springsfast.com/>

<http://www.asbg.com/>

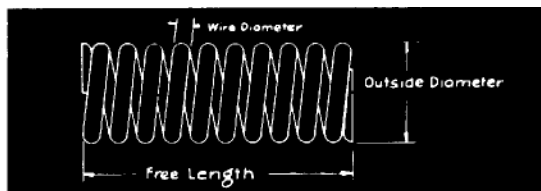
<http://www.centuryspring.com/>

COMPRESSION SPRINGS

FREE LENGTHS

Will go in hole In	7/16			1/2				5/8				3/4				7/8			
Wire Dia. In.	.031	.047	.062	.047	.062	.078	.094	.047	.062	.078	.094	.062	.078	.094	.125	.062	.078	.094	.125
Catalog No.	247	248	249																
Price Code	HB	HB	HC																
lbs./in.	12	55	180																
Max. Defl.	.32	.23	.16																
Catalog No.	250	251	252	283	284	285	286												
Price Code	HB	HB	HC	HB	HB	HE	HE												
lbs./in.	10	47	150	37	110	320	840												
Max. Defl.	.37	.27	.19	.29	.22	.15	.10												
Catalog No.	253	254	255	287	288	289	290	331	332	333	334								
Price Code	HB	HB	HC	HB	HB	HE	HE	HB	HB	HD	HE								
lbs./in.	7.9	36	175	29	85	240	610	20	54	140	320								
Max. Defl.	.47	.35	.25	.38	.29	.20	.14	.43	.35	.27	.20								
Catalog No.	256	257	258	291	292	293	294	335	336	337	338	375	376	377	378				
Price Code	HB	HB	HC	HB	HB	HE	HE	HB	HB	HD	HE	HD	HD	HF	HG				
lbs./in.	6.4	29	90	23	68	185	470	16	43	105	250	32	78	170	650				
Max. Defl.	.58	.44	.32	.48	.37	.26	.18	.53	.44	.34	.26	.59	.40	.32	.19				
Catalog No.	259	260	261	295	296	297	298	339	340	341	342	379	380	381	382	419	420	421	422
Price Code	HB	HB	HC	HB	HB	HE	HE	HB	HB	HD	HE	HD	HD	HF	HG	HF	HF	HJ	HK
lbs./in.	5.4	24	75	19	56	155	384	13	36	90	204	27	65	140	520	21	49	100	350
Max. Defl.	.68	.52	.39	.57	.44	.32	.23	.64	.53	.42	.32	.58	.48	.39	.24	.62	.53	.44	.30
Catalog No.	262	263	264	299	300	301	302	343	344	345	346	383	384	385	386	423	424	425	426
Price Code	HB	HB	HC	HB	HC	HE	HE	HB	HC	HD	HE	HD	HD	HF	HG	HF	HF	HJ	HK
lbs./in.	4.7	21	65	17	48	130	320	11	31	77	170	23	55	115	430	18	42	86	290
Max. Defl.	.79	.60	.45	.66	.51	.37	.27	.74	.62	.49	.38	.68	.57	.46	.29	.73	.63	.53	.36
Catalog No.	265	266	267	303	304	305	306	347	348	349	350	387	388	389	390	427	428	429	430
Price Code	HB	HC	HC	HC	HC	HF	HF	HC	HC	HD	HE	HE	HE	HF	HG	HG	HG	HJ	HK
lbs./in.	3.7	16	50	13	38	100	245	9.0	24	59	130	18	42	89	320	14	32	66	220
Max. Defl.	1.0	.77	.58	.84	.66	.49	.35	.94	.80	.64	.49	.88	.74	.60	.39	.94	.81	.69	.47
Catalog No.	268	269	270	307	308	309	310	351	352	353	354	391	392	393	394	431	432	433	434
Price Code	HB	HC	HC	HC	HC	HF	HF	HD	HD	HE	HF	HE	HE	HF	HG	HG	HG	HJ	HK
lbs./in.	3.1	14	41	11	31	83	200	7.4	20	48	105	15	34	72	260	12	26	53	175
Max. Defl.	1.2	.94	.70	1.0	.81	.60	.43	1.1	.98	.78	.61	1.1	.91	.74	.48	1.1	1.0	.85	.59
Catalog No.	271	272	273	311	312	313	314	355	356	357	358	395	396	397	398	435	436	437	438
Price Code	HC	HD	HE	HC	HD	HF	HF	HD	HD	HC	HF	HE	HE	HF	HG	HG	HG	HJ	HK
lbs./in.	2.6	11	35	9.1	26	70	170	6.2	17	41	90	12.4	29	61	216	9.9	22	45	147
Max. Defl.	1.4	1.1	.84	1.2	.96	.71	.52	1.3	1.1	.93	.73	1.3	1.1	.89	.58	1.35	1.2	1.0	.71
Catalog No.	274	275	276	315	316	317	318	359	360	361	362	399	400	401	402	439	440	441	442
Price Code	HD	HD	HE	HC	HD	HF	HF	HD	HD	HE	HF	HE	HE	HF	HG	HG	HG	HJ	HL
lbs./in.	2.3	10	30	7.9	23	60	145	5.4	14	35	77	11	25	52	185	8.6	19	38	125
Max. Defl.	1.6	1.3	.96	1.4	1.1	.82	.60	1.5	1.3	1.1	.85	1.4	1.2	1.0	.68	1.5	1.4	1.2	.83
Catalog No.	277	278	279	319	320	321	322	363	364	365	366	403	404	405	406	443	444	445	446
Price Code	HD	HE	HE	HF	HF	HG	HG	HD	HD	HF	HG	HF	HF	HF	HG	HJ	HJ	HK	HL
lbs./in.	1.5	6.6	20	5.2	15	39	94	3.6	9.4	23	50	7	16	34	115	5.6	12	25	80
Max. Defl.	2.4	1.9	1.4	2.1	1.7	1.2	.93	2.4	2.0	1.6	1.3	2.2	1.9	1.6	1.0	2.4	2.1	1.8	1.3
Catalog No.	280	281	282	323	324	325	326	367	368	369	370	407	408	409	410	447	448	449	450
Price Code	HE	HE	HE	HE	HF	HG	HG	HE	HE	HF	HG	HF	HF	HF	HL	HJ	HJ	HL	HM
lbs./in.	1.1	4.9	15	3.9	11	29	69	2.6	6.9	17	37	5.2	12	25	86	4.2	9.2	18	59
Max. Defl.	3.3	2.6	2.0	2.8	2.3	1.7	1.2	3.2	2.7	2.2	1.8	3.0	2.6	2.1	1.4	3.2	2.8	2.5	1.8
Catalog No.				327	328	329	330	371	372	373	374	411	412	413	414	451	452	453	454
Price Code				HF	HF	HG	HG	HF	HF	HF	HG	HG	HG	HJ	HN	HK	HK	HL	HO
lbs./in.				2.5	7.0	17	45	1.8	4.6	11	24	3.4	7.9	16	56	2.7	6.0	12	38
Max. Defl.				4.4	3.5	2.5	2	4.8	4.2	3.4	2.7	4.6	3.9	3.3	2.2	4.9	4.3	3.8	2.7
Catalog No.												415	416	417	418	455	456	457	458
Price Code												HJ	HJ	HK	HO	HL	HL	HM	HP
lbs./in.												2.6	6	11	40	2.0	4.5	8.9	28
Max. Defl.												6.1	5.2	4.5	3.0	6.5	5.8	5.1	3.7
Maximum Load	3.7	12.7	29	11	25	45	88	8.3	19	38	66	15.8	31.2	54	125	13.4	26.3	45	105
Will work free over	.347	.315	.285	.375	.345	.313	.281	.505	.475	.443	.411	.585	.554	.522	.460	.700	.670	.638	.576
Pitch	.195	.141	.128	.173	.151	.141	.141	.259	.214	.188	.177	.284	.240	.217	.204	.371	.306	.268	.239
Solid Stress (000 omitted)	125	118	113	118	113	109	105	118	113	109	105	113	109	105	99	113	109	105	99

NOTE: Stock springs can be ordered in stainless steel or plated. Prices quoted upon request.



FREE LENGTHS	Will go in hole In.		1				1 1/4			1 1/2			2			3			4			6	
	Wire Dia.	In.	.078	.094	.125	.187	.094	.125	.187	.125	.187	.250	.187	.250	.375	.250	.375	.500	.375	.500	.750	.750	1.000
1	Catalog No.		459	460	461	462																	
	Price Code		HK	HL	HL	HR																	
	lbs./in.		34	67	210	1500																	
	Max. Defl.		.67	.58	.41	.19																	
1 1/4	Catalog No.		463	464	465	466	499	500	501														
	Price Code		HL	HM	HM	HS	HN	HN	HM														
	lbs./in.		26	52	160	1100	35	100	600														
	Max. Defl.		.87	.76	.55	.26	.85	.67	.37														
1 1/2	Catalog No.		467	468	469	470	502	503	504	526	527	528											
	Price Code		HL	HM	HN	HS	HN	HO	HT	HR	HX	HAC											
	lbs./in.		21	42	130	870	29	82	460	60	300	1200											
	Max. Defl.		1.0	.93	.69	.34	1.1	.84	.48	.95	.60	.35											
2	Catalog No.		471	472	473	474	505	506	507	529	530	531											
	Price Code		HL	HM	HN	HS	HN	HO	HT	HR	HX	HAC											
	lbs./in.		18	35	108	712	24	68	379	50	244	960											
	Max. Defl.		1.3	1.1	.83	.41	1.3	1.0	.59	1.1	.74	.44											
3	Catalog No.		475	476	477	478	508	509	510	532	533	534	553	554	555								
	Price Code		HM	HN	HO	HT	HO	HP	HU	HS	HZ	HAE	HAA	HAG	HZZ								
	lbs./in.		16	30	93	600	21	59	320	43	200	800	115	390	3000								
	Max. Defl.		1.4	1.3	.97	.49	1.4	1.2	.70	1.3	.87	.53	1.1	.77	.34								
4	Catalog No.		479	480	481	482	511	512	513	535	536	537	556	557	558	577	578	579					
	Price Code		HN	HO	HP	HZ	HP	HR	HAA	HT	HAD	HAL	HAE	HAN	HZZ	HAR	HZZ	HZZ					
	lbs./in.		10	19	59	370	13	37	200	27	130	480	73	230	1650	105	560	2300					
	Max. Defl.		2.2	2.0	1.5	.79	2.2	1.8	1.1	2.1	1.4	.89	1.8	1.3	.61	1.8	1.1	.64					
6	Catalog No.		483	484	485	486	514	515	516	538	539	540	559	560	561	580	581	582	598	599	610		
	Price Code		HP	HR	HS	HAC	HS	HT	HAD	HW	HAG	HAO	HAJ	HAR	HZZ	HAT	HZZ	HZZ	HZZ	HZZ	HZZ		
	lbs./in.		7.4	14	43	270	9.9	27	144	20	93	340	53	170	1150	76	390	1500	210	720	4600		
	Max. Defl.		3.0	2.7	2.1	1.1	3.0	2.5	1.5	2.8	1.9	1.2	2.5	1.8	.88	2.5	1.6	.96	2.1	1.4	.8		
8	Catalog No.		487	488	489	490	517	518	519	541	542	543	562	563	564	583	584	585	600	601	611	616	621
	Price Code		HR	HT	HU	HAD	HU	HW	HAE	HX	HAI	HAT	HAM	HAU	HZZ	HAI	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ
	lbs./in.		4.9	9.4	28	175	6.5	18	93	13	60	220	34	105	710	49	240	920	130	430	2840	850	3500
	Max. Defl.		4.6	4.1	3.2	1.7	4.7	3.9	2.4	4.3	3.0	1.9	3.8	2.8	1.4	4.0	2.6	1.6	3.4	2.4	1.3	1.9	1.4
12	Catalog No.		491	492	493	494	520	521	522	544	545	546	565	566	567	586	587	588	602	603	612	617	622
	Price Code		HS	HU	HW	HAL	HW	HX	HAM	HAA	HAP	HAW	HAR	HAZ	HZZ	HBD	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ
	lbs./in.		3.6	7.0	21	125	4.8	13	68	9.6	44	160	25	79	510	36	175	660	95	310	2050	630	2500
	Max. Defl.		6.2	5.6	4.3	2.3	6.3	5.2	3.2	5.9	4.1	2.7	5.2	3.9	1.9	5.4	3.6	2.2	4.5	3.4	1.8	2.7	2.0
16	Catalog No.		495	496	497	498	523	524	525	547	548	549	568	569	570	589	590	591	604	605	613	618	623
	Price Code		HT	HW	HZ	HAP	HX	HAA	HAR	HAC	HAU	HBA	HAZ	HBE	HZZ	HBK	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ
	lbs./in.		2.4	4.6	14	84	3.2	8.7	45	6.3	29	105	16	52	330	23	110	420	61	195	1325	400	1580
	Max. Defl.		9.4	8.4	6.5	3.5	9.5	7.9	5.0	8.9	6.2	4.1	8.0	5.9	3.0	8.3	5.5	3.5	7.3	5.3	2.6	4.3	3.1
24	Catalog No.									550	551	552	571	572	573	592	593	594	606	607	614	619	624
	Price Code									HAE	HAW	HBD	HAZ	HBG	HZZ	HBL	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ	HZZ
	lbs./in.									4.7	21	74	12	38	240	17	83	310	45	145	975	300	1170
	Max. Defl.									11.9	8.5	5.6	10.7	8.0	4.1	11.3	7.5	4.0	10	7.3	3.8	6.1	4.3
Maximum Load			23	39	90	295	30	69	224	57	180	428	131	307	1000	195	624	1470	449	1040	3700	2000	4800
Will work free over			784	752	690	565	100	940	815	1.19	1.06	.940	1.52	1.39	1.14	2.33	2.08	1.83	3.08	2.83	2.25	4.25	3.75
Pitch			382	328	279	268	481	384	327	516	403	388	596	518	514	917	741	736	108	969	100	1.3	1.4
Solid Stress (1000 omitted)			109	105	99	90	105	99	90	99	90	85	90	85	77	85	77	73	77	73	70	70	65



Hardware Products Company, Inc.

EXTENSION SPRINGS



ORDER BY:
SE LENGTH x O.D. x WIRE DIA.

SPECIFY HOOKS OR LOOPS

The figures given for "Maximum Extension" and "lbs. per inch" are for a spring 1" long. For other lengths multiply the "Maximum Extension" and divide the "lbs. per inch" by the length in inches. The "Maximum Load" and "Initial Tension" remain constant for any length.

Example: A spring 1/2" diam. .062" wire and 4" long will have a safe maximum extension of 3.2" and it will require 4 lbs. to deflect it 1 in. The spring will hold approximately 3.3 lbs. before it starts to extend, and will hold a maximum of 16.1 lbs. without permanent stretch. If 8.5 lbs. is hung on the spring it will deflect 1.3". 8.5 lbs. minus 3.3 lbs. divided by 4 lbs. per inch equals 1.3".

NOTE: Stock springs can be ordered in stainless steel or plated. Prices quoted upon request.

Outside dia.	Wire dia.	Catalog No.	Price code	Safe maximum load in pounds	Safe maximum extension in.	Approx. initial tension in pounds	Pound per inch extension	Stress at max. load (1000 omitted)	Weight per foot (lbs.)
1/8	.012	01	EHE	.6	1.9	.07	.27	100	.012
	.016	02	EHD	1.3	.9	.2	1.2	93	.015
	.023	03	EHD	4.2	.35	.9	9.0	90	.02
5/32	.012	04	EHE	.47	3.5	.01	.12	100	.015
	.016	05	EHD	1.1	1.7	.15	.55	93	.019
	.023	06	EHD	3.2	.7	.5	3.9	90	.027
3/16	.016	07	EHD	.87	2.5	.1	.3	93	.024
	.023	08	EHD	2.6	1.0	.4	2.2	90	.032
	.031	09	EHD	6.5	.45	1.5	10.7	88	.04
7/32	.016	10	EHD	.75	4.0	.01	.18	93	.028
	.023	11	EHD	2.3	1.6	.32	1.2	90	.039
	.031	12	EHD	5.5	.7	1.0	6.5	88	.048
1/4	.023	13	EHD	1.8	1.9	.26	.8	90	.044
	.031	14	EHD	4.7	1.0	.75	3.8	88	.055
	.047	15	EHE	16.0	.3	3.5	40.0	83	.082
5/16	.023	16	EHE	1.5	3.5	.16	.38	90	.058
	.031	17	EHE	3.6	1.6	.55	1.9	88	.072
	.047	18	EHE	12.5	.9	2.2	10.8	83	.108
3/8	.031	19	EHE	2.9	2.5	.37	1.0	88	.084
	.047	20	EHE	10.5	.9	1.7	9.5	83	.13
	.062	21	EHF	23.0	.39	5.3	45.0	79	.16
7/16	.031	22	EHF	2.5	3.5	.26	.63	88	.105
	.047	23	EHF	8.5	1.2	1.4	5.7	83	.163
	.062	24	EHF	20.0	.6	4.3	26.0	79	.2
1/2	.047	25	EHF	7.3	1.6	1.1	3.7	83	.18
	.062	26	EHF	17.0	.8	3.3	16.0	79	.23
	.078	27	EHG	34.0	.45	8.0	57.0	77	.28
5/8	.094	28	EHJ	57.0	.25	16.0	160.0	74	.32
	.047	29	EHG	6.0	3.0	.7	1.7	83	.24
	.062	30	EHG	13.3	1.4	2.1	7.6	79	.3
3/4	.078	31	EHG	27.0	.9	5.2	23.0	77	.37
	.094	32	EHJ	45.0	.4	11.0	73.0	74	.44
	.062	33	EHG	10.5	2.2	1.5	4.1	79	.36
7/8	.078	34	EHJ	22.0	1.3	3.5	14.0	77	.46
	.094	35	EHJ	36.0	.7	8.0	38.0	74	.51
	.125	36	EHK	85.0	.3	22.0	180.0	69	.64
1	.062	37	EHK	9.2	3.3	1.1	2.4	79	.4
	.078	38	EHL	18.0	1.7	2.6	8.7	77	.59
	.094	39	EHL	31.0	1.0	6.0	25.0	74	.64
1 1/4	.125	40	EHL	72.0	.5	17.0	107.0	69	.8
	.078	41	EHL	16.0	2.5	2.0	5.5	77	.67
	.094	42	EHL	26.0	1.5	4.5	13.7	74	.70
1 1/2	.125	43	EHL	85.0	.75	14.0	68.0	69	.90
	.187	44	EHW	200.0	.23	60.0	600.0	63	1.4
	.094	45	EHL	21.0	2.6	2.8	6.8	74	.94
2	.125	46	EHL	47.0	1.2	9.0	31.0	69	1.3
	.187	47	EHL	148.0	.3	40.0	290.0	63	1.8
	.125	48	EHL	39.0	1.9	6.0	17.0	69	1.4
2 1/2	.187	49	EHL	122.0	.6	33.0	150.0	63	2.2
	.250	50	EHL	290.0	.27	90.0	720.0	60	2.6
	.187	51	EHL	90.0	1.3	20.0	54.0	63	3.1
3	.250	52	EHL	210.0	.6	55.0	260.0	60	3.7

Carried in stock in 3-foot lengths - cut to length and looped to order.

Hardware Products Company, Inc.

191 WILLIAMS STREET • CHELSEA, MA 02150

Appendix E

COUPLER CURVE ATLASES

E.1 HRONES AND NELSON ATLAS OF FOURBAR LINKAGES

The entire Hrones and Nelson coupler curve atlas is downloadable as PDF files. Figure 3-17 in Section 3.6 shows one page from this atlas and describes how to use it. Read the first chapter within the Hrones and Nelson atlas for more information on how it is arranged and how to use it. The downloadable video [*Coupler Curves and Linkage Atlases*](#) gives detailed instructions on its use and shows an example. Once you extract a trial linkage geometry from the atlas, use program **LINKAGES** to investigate its behavior and to vary the linkage geometry.

E.2 ZHANG, NORTON, HAMMOND ATLAS OF GEARED FIVEBAR LINKAGES

The entire Zhang atlas is downloadable as PDF files. A sample page is shown in Figure 3-23. Read the first chapter within the Zhang atlas for information on how it is arranged and how to use it. See Sections 2.4, 3.6, 4.9, 6.8, and 7.4 for more information on the geared fivebar linkages. The video [*Coupler Curves and Linkage Atlases*](#) gives a brief overview of this atlas. Once you extract a trial linkage geometry from the atlas, use program **LINKAGES** to investigate its behavior and to vary the linkage geometry.

A summary of the parameters in the Zhang atlas is:

$\alpha = \text{Coupler Link 3 / Link 2}$

$\beta = \text{Ground Link 1 / Link 2}$

$\lambda = \text{Gear Ratio} = \text{Gear 5 / Gear 2}.$

Phase angle is noted on each plot of a coupler curve.

The dots along curves are at every 10 degrees of Link 2's rotation.

Linkage is symmetrical: Link 2 = Link 5 and Link 3 = Link 4

Note that the λ in the atlas is the inverse of the λ that is defined in Sections 4.9, 6.8, and 7.4. See also Figures P4-4, P6-4, and P7-4. For example, a gear ratio λ of 2 in the Zhang atlas corresponds to a λ of 0.5 in the text and in program **LINKAGES**. (The difference merely corresponds to a mirroring of the linkage from left to right.)

Appendix F

ANSWERS TO SELECTED PROBLEMS

CHAPTER 2 KINEMATICS FUNDAMENTALS

2-1

- | | | | | | | |
|------|------|------|------|------|------|------|
| a. 1 | b. 1 | c. 2 | d. 1 | e. 7 | f. 1 | g. 4 |
| h. 4 | i. 4 | j. 2 | k. 1 | l. 1 | m. 2 | n. 2 |
| o. 4 | | | | | | |

- 2-3** a. 1 b. 3 c. 3 d. 3 e. 2

- 2-4** a. 6 b. 6 c. 5 d. 4, but 2 are dynamically coupled* e. 4 f. 3

- 2-5** force-closed

2-6

- a. pure rotation
- b. complex planar motion
- c. pure translation
- d. pure translation
- e. pure rotation
- f. complex planar motion
- g. complex planar motion

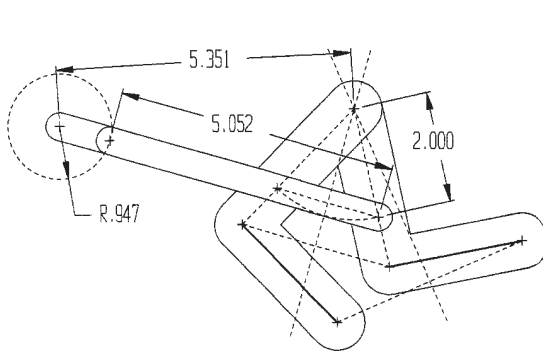
- 2-7** a. 0 b. 1 c. 1 d. 3

2-8

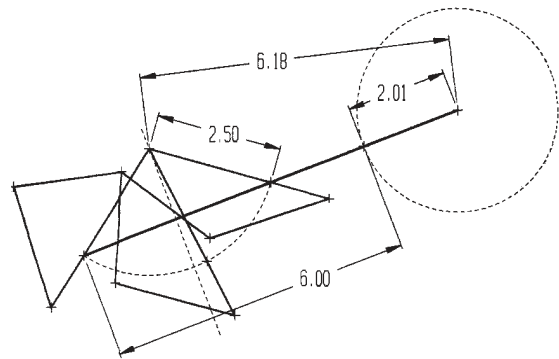
- a. structure - $DOF = 0$
- b. mechanism - $DOF = 1$
- c. mechanism - $DOF = 1$
- d. mechanism - $DOF = 3$

- 2-15** a. Grashof b. non-Grashof c. special-case Grashof

* Dynamically coupled means that, at speed, leaning the bike to the side results in its turning to the side to which it is leaning. So the angular freedom of this machine in the plane of the road is coupled with its ability to rotate about its long axis (lean). Except at very low speed, you steer a motorcycle by pushing down (toward the ground) on the handlebar on the inside of the turn, rather than by actually turning the handlebar in the direction of the turn. If you are moving the bike with your feet to park it, then you turn the handlebar. But at any significant speed, the gyroscopic effect takes over and leaning the bike makes it turn. This is true of a pedal bike as well if it has sufficient forward speed.



(a) One possible solution to Problem 3-3



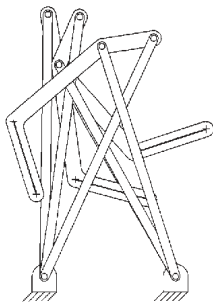
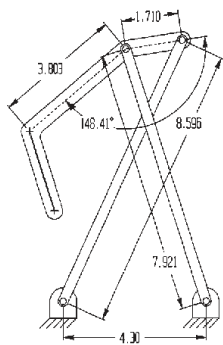
(b) One possible solution to Problem 3-5

FIGURE S3-1

Solutions to Problems 3-3 and 3-5

2-21

- | | | |
|------------|-------------------------|------------|
| a. $M = 1$ | b. $M = 1$ | c. $M = 1$ |
| d. $M = 1$ | e. $M = -1$ (a paradox) | f. $M = 1$ |
| g. $M = 1$ | h. $M = 0$ (a paradox) | |

2-24 a. $M = 1$ b. $M = 1$ **2-26** $M = 1$ **2-27** $M = 0$ **2-35** $M = 1$, fourbar slider-crank**2-61** a. $M = 3$ b. $M = 2$ c. $M = 1$ **2-62** a. $M = 1$ b. $M = 2$ c. $M = 4$ **FIGURE S3-2**

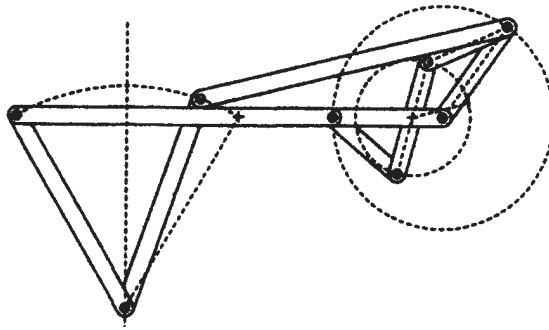
Unique solution to Problem 3-6

CHAPTER 3**GRAPHICAL LINKAGE SYNTHESIS****3-1**

- path generation
- motion generation
- function generation
- path generation
- path generation

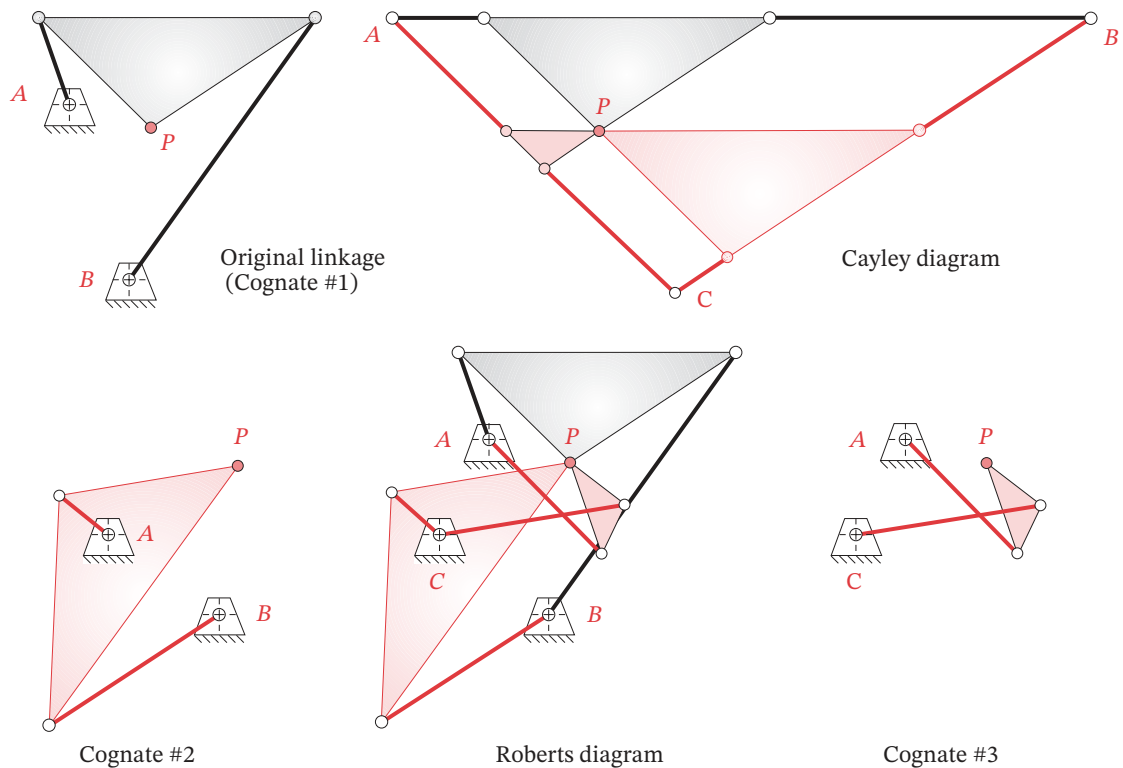
Note that synthesis problems have many valid solutions. We cannot provide a “right answer” to all of these design problems. Check your solution with a cardboard model and/or by putting it into one of the programs supplied with the text.

3-3 See Figure S3-1.**3-5** See Figure S3-1.**3-6** See Figure S3-2.**3-8** See Figure S3-3.

**FIGURE S3-3**

One possible solution to Problem 3-8

- 3-10** The solution using Figure 3-17 is shown in Figure S3-4. (Use program LINKAGES to check your solution.)
- 3-22** The transmission angle ranges from 31.5° to 89.9° .
- 3-23** Grashof crank-rocker. Transmission angle ranges from 58.1° to 89.8° .

**FIGURE S3-4**

Solution to Problem 3-10. Finding the cognates of the fourbar linkage shown in Figure 3-17

- 3-31** $L_1 = 160.6$, $L_2 = 81.3$, $L_3 = 200.2$, $L_4 = 200.2$ mm.
- 3-36** Grashof double-rocker. Works from 56° to 158° and from 202° to 310° . Transmission angle ranges from 0° to 90° .
- 3-39** Non-Grashof triple-rocker. Toggles at $\pm 116^\circ$. Transmission angle ranges from 0° to 88° .
- 3-42** Non-Grashof triple-rocker. Toggles at $\pm 55.4^\circ$. Transmission angle ranges from 0° to 88.8° .
- 3-79** Link 2 = 1, link 3 = link 4 = link 1 = 1.5. Coupler point is at 1.414 @ 135° versus link 3. Put these data into program LINKAGES to see the coupler curve.

CHAPTER 4 POSITION ANALYSIS

- 4-6** and **4-7** See Table S4-1 and the file P07-04row.4br.
- 4-9** and **4-10** See Table S4-2.
- 4-11** and **4-12** See Table S4-3.
- 4-13** See Table S4-1.
- 4-14** Open the file P07-04row.4br[†] in program LINKAGES to see this solution.*
- 4-15** Open the file P07-04row.4br[†] in program LINKAGES to see this solution.*
- 4-16** See Table S4-4.
- 4-17** See Table S4-4.
- 4-21** Open the file P04-21.4br in program LINKAGES to see this solution.*
- 4-23** Open the file P04-23.4br in program LINKAGES to see this solution.*
- 4-25** Open the file P04-25.4br in program LINKAGES to see this solution.*
- 4-26** Open the file P04-26.4br in program LINKAGES to see this solution.*
- 4-29** Open the file P04-29.4br in program LINKAGES to see this solution.*
- 4-30** Open the file P04-30.4br in program LINKAGES to see this solution.*
- 4-31** $r_1 = -6.265$, $r_2 = -0.709$.

CHAPTER 5 ANALYTICAL LINKAGE SYNTHESIS

- 5-8** Given: $\alpha_2 = -62.5^\circ$, $P_{21} = 2.47$, $\delta_2 = 120^\circ$
 For left dyad: Assume: $z = 1.075$, $\phi = 204^\circ$ $\beta_2 = -27^\circ$
 Calculate: $W = 3.67$ @ -113.5°
 For right dyad: Assume: $s = 1.24$, $\psi = 74^\circ$ $\gamma_2 = -40^\circ$
 Calculate: $U = 5.46$ @ -125.6°

* These files can be found in the PROBLEM SOLUTIONS folder downloadable with this text.

[†] The letter x in the filename represents the row number from the table of problem data.

TABLE S4-1 Solutions for Problems 4-6, 4-7, and 4-13

Row	θ_3 Open	θ_4 Open	Trans Ang	θ_3 Crossed	θ_4 Crossed	Trans Ang
<i>a</i>	88.8	117.3	28.4	-115.2	-143.6	28.4
<i>c</i>	-53.1	16.5	69.6	173.3	103.6	69.6
<i>e</i>	7.5	78.2	70.7	-79.0	-149.7	70.7
<i>g</i>	-16.3	7.2	23.5	155.7	132.2	23.5
<i>i</i>	-1.5	103.1	75.4	-113.5	141.8	75.4
<i>k</i>	-13.2	31.9	45.2	-102.1	-147.3	45.2
<i>m</i>	-3.5	35.9	39.4	-96.5	-135.9	39.4

TABLE S4-2 Solutions for Problems 4-9 to 4-10

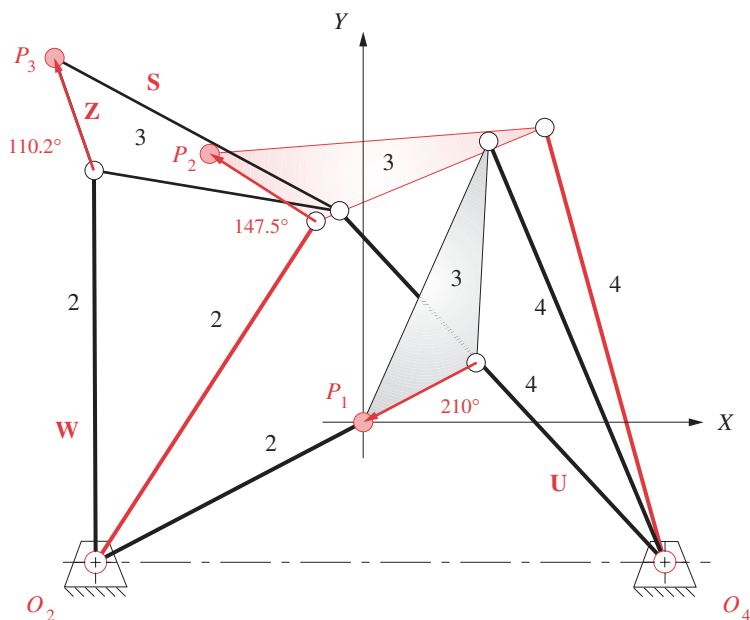
Row	θ_3 Open	Slider Open	θ_3 Crossed	Slider Crossed
<i>a</i>	180.1	5.0	-0.14	-3.0
<i>c</i>	205.9	9.8	-25.90	-4.6
<i>e</i>	175.0	16.4	4.20	-23.5
<i>g</i>	212.7	27.1	-32.70	-14.9

TABLE S4-3 Solutions for Problems 4-11 to 4-12

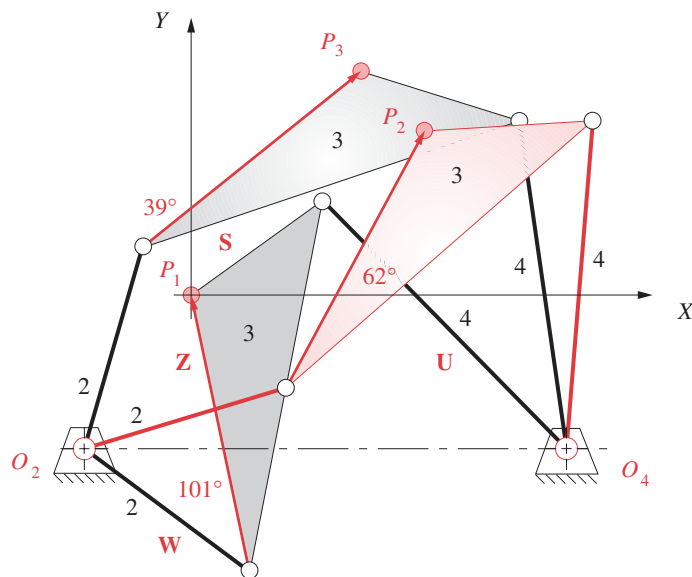
Row	θ_3 Open	θ_4 Open	R_B Open	θ_3 Crossed	θ_4 Crossed	R_B Crossed
<i>a</i>	232.7	142.7	1.79	-79.0	-169.0	1.79
<i>c</i>	91.4	46.4	2.72	208.7	163.7	11.20
<i>e</i>	158.2	128.2	6.17	-36.2	-66.2	9.63

TABLE S4-4 Solutions for Problems 4-16 to 4-17

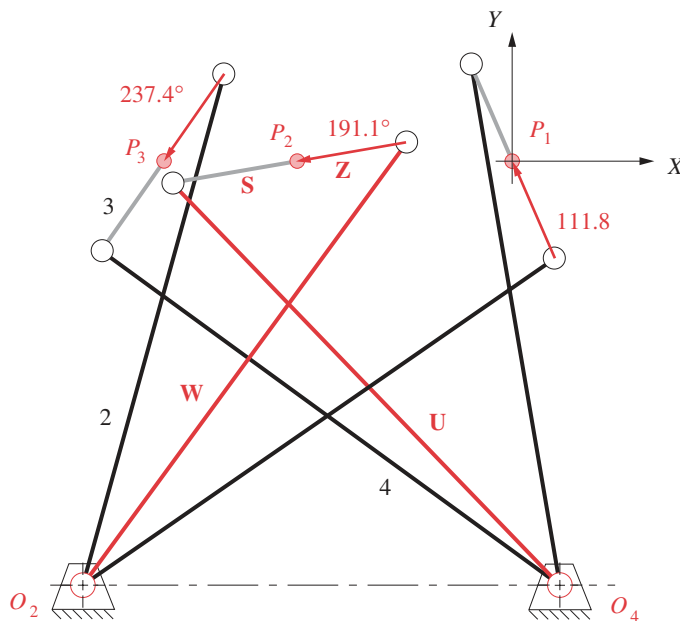
Row	θ_3 Open	θ_4 Open	θ_3 Crossed	θ_4 Crossed
<i>a</i>	173.6	-177.7	-115.2	-124.0
<i>c</i>	17.6	64.0	-133.7	180.0
<i>e</i>	-164.0	-94.4	111.2	41.6
<i>g</i>	44.2	124.4	-69.1	-149.3
<i>i</i>	37.1	120.2	-67.4	-150.5

**FIGURE S5-1**

Solution to Problem 5-11. Open the file P05-11 in program LINKAGES for more information

**FIGURE S5-2**

Solution to Problem 5-15. Open the file P05-15 in program LINKAGES for more information

**FIGURE S5-3**

Solution to Problem 5-19. Open the file P05-19 in program LINKAGES for more information

5-11 See Figure S5-1 for the solution. The link lengths are:

$$\text{Link 1} = 4.35, \quad \text{Link 2} = 3.39, \quad \text{Link 3} = 1.94, \quad \text{Link 4} = 3.87$$

5-15 See Figure S5-2 for the solution. The link lengths are:

$$\text{Link 1} = 3.95, \quad \text{Link 2} = 1.68, \quad \text{Link 3} = 3.05, \quad \text{Link 4} = 0.89$$

5-19 See Figure S5-3 for the solution. The link lengths are:

$$\text{Link 1} = 2, \quad \text{Link 2} = 2.5, \quad \text{Link 3} = 1, \quad \text{Link 4} = 2.5$$

5-26 Given:

$$\begin{aligned} \alpha_2 &= -45^\circ, & P_{21} &= 184.78 \text{ mm}, & \delta_2 &= -5.28^\circ \\ \alpha_3 &= -90^\circ, & P_{31} &= 277.35 \text{ mm}, & \delta_3 &= -40.47^\circ \\ O_{2x} &= 86 \text{ mm} & O_{2y} &= -132 \text{ mm} \\ O_{4x} &= 104 \text{ mm} & O_{4y} &= -155 \text{ mm} \end{aligned}$$

For left dyad: Calculate: $\beta_2 = -85.24^\circ$ $\beta_3 = -164.47^\circ$

Calculate: $W = 110.88 \text{ mm}$ $\theta = 124.24^\circ$

Calculate: $Z = 46.74 \text{ mm}$ $\phi = 120.34^\circ$

For right dyad: Calculate: $\gamma_2 = -75.25^\circ$ $\gamma_3 = -159.53^\circ$

Calculate: $U = 120.70 \text{ mm}$ $\sigma = 104.35^\circ$

Calculate: $S = 83.29 \text{ mm}$ $\psi = 152.80^\circ$

5-33	Given:	$\alpha_2 = -25^\circ$,	$P_{21} = 133.20 \text{ mm}$,	$\delta_2 = -12.58^\circ$
		$\alpha_3 = -101^\circ$,	$P_{31} = 238.48 \text{ mm}$,	$\delta_3 = -51.64^\circ$
		$O_{2x} = -6.2 \text{ mm}$	$O_{2y} = -164.0 \text{ mm}$	
		$O_{4x} = 28.0 \text{ mm}$	$O_{4y} = -121.0 \text{ mm}$	
	For left dyad:	Calculate: $\beta_2 = -53.07^\circ$	$\beta_3 = -94.11^\circ$	
		Calculate: $W = 128.34 \text{ mm}$	$\theta = 118.85^\circ$	
		Calculate: $Z = 85.45 \text{ mm}$	$\phi = 37.14^\circ$	
	For right dyad:	Calculate: $\gamma_2 = -77.26^\circ$	$\gamma_3 = -145.66^\circ$	
		Calculate: $U = 92.80 \text{ mm}$	$\sigma = 119.98^\circ$	
		Calculate: $S = 83.29 \text{ mm}$	$\psi = 65.66^\circ$	
5-35	Given:	$\alpha_2 = -29.4^\circ$,	$P_{21} = 99.85 \text{ mm}$,	$\delta_2 = 7.48^\circ$
		$\alpha_3 = -2.3^\circ$,	$P_{31} = 188.23 \text{ mm}$,	$\delta_3 = -53.75^\circ$
		$O_{2x} = -111.5 \text{ mm}$	$O_{2y} = 183.2 \text{ mm}$	
		$O_{4x} = -111.5 \text{ mm}$	$O_{4y} = -38.8 \text{ mm}$	
	For left dyad:	Calculate: $\beta_2 = 69.98^\circ$	$\beta_3 = 139.91^\circ$	
		Calculate: $W = 100.06 \text{ mm}$	$\theta = 150.03^\circ$	
		Calculate: $Z = 306.82 \text{ mm}$	$\phi = -49.64^\circ$	
	For right dyad:	Calculate: $\gamma_2 = -4.95^\circ$	$\gamma_3 = -48.81^\circ$	
		Calculate: $U = 232.66 \text{ mm}$	$\sigma = 62.27^\circ$	
		Calculate: $S = 167.17 \text{ mm}$	$\psi = -88.89^\circ$	

CHAPTER 6 VELOCITY ANALYSIS

- 6-4** and **6-5** See Table S6-1 and the file P07-04row.4br.
- 6-6** and **6-7** See Table S6-2.
- 6-8** and **6-9** See Table S6-3.
- 6-10** and **6-11** See Table S6-4.
- 6-16** $V_A = 12 \text{ in/sec @ } 124.3^\circ$, $V_B = 11.5 \text{ in/sec @ } 180^\circ$, $V_C = 5.65 \text{ in/sec @ } 153.3^\circ$, $\omega_3 = -5.69 \text{ rad/sec}$.
- 6-47** Open the file P06-47.4br in program LINKAGES to see this solution.*
- 6-48** Open the file P06-48.4br in program LINKAGES to see this solution.*
- 6-49** Open the file P06-49.4br in program LINKAGES to see this solution.*
- 6-51** Open the file P06-51.4br in program LINKAGES to see this solution.*
- 6-62** Open the file P06-62.4br in program LINKAGES to see this solution.*
- 6-65** $V_A = 94.5 \text{ in/sec}$, $V_B = 115.2$, $V_{slip} = 162.8$, $V = 65.9$, $\omega = -70 \text{ rad/sec}$.

* These files can be found in the PROBLEM SOLUTIONS folder downloadable with this text.

TABLE S6-1 Solutions for Problems 6-4 to 6-5

Row	ω_3 Open	ω_4 Open	V_P Mag	V_P Ang	ω_3 Crossed	ω_4 Crossed	V_P Mag	V_P Ang
<i>a</i>	-6.0	-4.0	40.8	58.2	-0.66	-2.66	22.0	129.4
<i>c</i>	-12.7	-19.8	273.8	-53.3	-22.70	-15.70	119.1	199.9
<i>e</i>	1.85	-40.8	260.5	-12.1	-23.30	19.30	139.9	42.0
<i>g</i>	76.4	146.8	798.4	92.9	239.00	168.60	1435.3	153.9
<i>i</i>	-25.3	25.6	103.1	-13.4	56.90	6.00	476.5	70.4
<i>k</i>	-56.2	-94.8	436.0	-77.4	-55.60	-16.90	362.7	79.3
<i>m</i>	18.3	83.0	680.8	149.2	7.73	-57.00	571.3	133.5

TABLE S6-2 Solutions for Problems 6-6 to 6-7

Row	V_A Mag	V_A Ang	ω_3 Open	V_B Mag Open	ω_3 Crossed	V_B Mag Crossed
<i>a</i>	14	135	-2.47	-9.9	2.47	-9.92
<i>c</i>	45	-120	5.42	-41.5	-5.42	-3.54
<i>e</i>	250	135	-8.86	-189.7	8.86	-163.80
<i>g</i>	700	60	-28.80	738.9	28.80	-38.90

TABLE S6-3 Solutions for Problems 6-8 to 6-9

Row	V_A Mag	V_A Ang	ω_3 Open	V_{slip} Open	V_B mag Open	ω_3 Crossed	V_{slip} Crossed	V_B Mag Crossed
<i>a</i>	20.0	120.0	-10.3	33.5	41.2	-3.6	-4.25	14.6
<i>c</i>	240.0	135.0	23.7	73.0	142.5	-14.9	130.50	89.4
<i>e</i>	180.0	-15.0	-2.7	-176.0	5.4	5.7	162.00	11.5

TABLE S6-4 Solutions for Problems 6-10 to 6-11

Row	ω_3 Open	ω_4 Open	ω_3 Crossed	ω_4 Crossed
<i>a</i>	32.6	16.9	-75.2	-59.6
<i>c</i>	10.7	-2.6	-8.2	5.1
<i>e</i>	-158.3	-81.3	-116.8	-193.9
<i>g</i>	-8.9	-40.9	-48.5	-16.5
<i>i</i>	-40.1	47.9	59.6	-28.4

TABLE S7-1 Solutions for Problems 7-3 to 7-4

Row	α_3 Open	α_4 Open	A_P Mag	A_P Ang	α_3 Crossed	α_4 Crossed	A_P Mag	A_P Ang
a	26.1	53.3	419	240.4	77.9	50.7	298	-11.3
c	-154.4	-71.6	4400	238.9	-65.2	-148.0	3554	100.6
e	331.9	275.6	10 260	264.8	1287.7	1344.1	19 340	-65.5
g	-23 510.0	-19 783.0	172 688	191.0	-43 709.0	-47 436.0	273 634	-63.0
i	-344.6	505.3	9492	-81.1	121.9	-728.0	27 871	150.0
k	-2693.0	-4054.0	56 271	220.2	311.0	1672.1	27 759	-39.1
m	680.8	149.2	35 149	261.5	9266.1	10 303.0	63 831	103.9

TABLE S7-2 Solutions for Problems 7-5 to 7-6

Row	A_A Mag	A_A Ang	α_3 Open	A_B Mag Open	A_B Ang Open	α_3 Crossed	A_B Mag Crossed	A_B Ang Crossed
a	140	-135	25	124	180	-25	74	180
c	676	153	-29	709	180	29	490	180
e	12 500	45	-447	6653	0	447	11 095	0
g	70 000	150	-1136	62 688	180	1136	58 429	180

TABLE S7-3 Solutions for Problems 7-7 to 7-8

Row	α_3 Open	α_4 Open	A_{slip} Open	α_3 Crossed	α_4 Crossed	A_{slip} Crossed
a	130.5	130.5	-128.5	-9.9	-9.9	19.0
c	-212.9	-212.9	1078.8	-217.8	-217.8	-728.2
e	896.3	896.3	-1818.6	595.6	595.6	1822.6

TABLE S7-4 Solutions for Problem 7-9

Row	α_3 Open	α_4 Open	α_3 Crossed	α_4 Crossed
a	3191	2492	-6648	-5949
c	314	228	87	147
e	2171	-6524	7 781	5414
g	-22 064	-23 717	-5529	-29 133
i	-5697	-3380	-2593	-7184

CHAPTER 7 ACCELERATION ANALYSIS

- 7-3 and 7-4 See Table S7-1 and the file P07-04row.4br.
- 7-5 and 7-6 See Table S7-2.
- 7-7 and 7-8 See Table S7-3.
- 7-9 See Table S7-4.
- 7-12 176.9 in/sec^2 .
- 7-21 $A_A = 26.26 \text{ m/sec}^2 @ 211.1^\circ, A_B = 8.328 \text{ m/sec}^2 @ -13.9^\circ$.
- 7-24 $A_A = 16 \text{ m/sec}^2 @ 237.6^\circ, A_B = 12.01 \text{ m/sec}^2 @ 207.4^\circ, \alpha_4 = 92 \text{ rad/sec}^2$.
- 7-28 $A_A = 39.38 \text{ m/sec}^2 @ -129^\circ, A_B = 39.7 \text{ m/sec}^2 @ -90^\circ$.
- 7-39 Open the file P07-39.4br in program LINKAGES to see this solution.*
- 7-40 Open the file P07-40.4br in program LINKAGES to see this solution.*
- 7-41 Open the file P07-41.4br in program LINKAGES to see this solution.*
- 7-42 Open the file P07-42.4br in program LINKAGES to see this solution.*
- 7-44 Open the file P07-44.4br in program LINKAGES to see this solution.*
- 7-56 Tipover at 19.0 to 20.3 mph; load slides at 16.2 to 19.5 mph.
- 7-76 $A_D = 7\,554.1 \text{ in/sec}^2 @ 150.8^\circ, \alpha_6 = 692.98 \text{ rad/sec}^2$.
- 7-78 $A_A = 677.1 \text{ in/sec}^2 @ -119.7^\circ, A_B = 1\,337.5 \text{ in/sec}^2 @ -26.09^\circ, A_P = 730.37 \text{ in/sec}^2 @ -53.65^\circ, \alpha_4 = 431.175 \text{ rad/sec}^2$
- 7-87 $A_C = 37.5 \text{ in/sec}^2 @ 90^\circ$

CHAPTER 8 CAM DESIGN

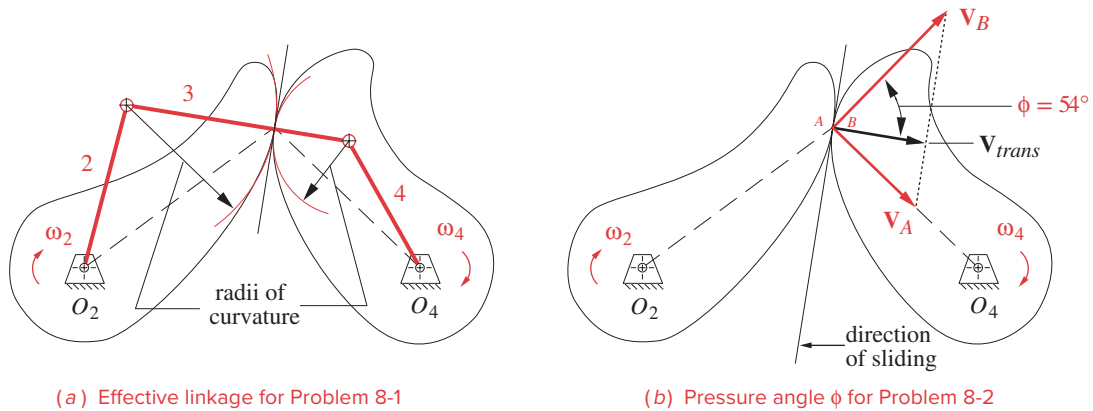
Most of the problems in this cam chapter are design problems with more than one correct solution. Use program DYNACAM to check your solution obtained with *Mathcad*, *Matlab*, *Excel*, or *TKSolver* and also to explore various solutions and compare them to find the best one for the constraints given in each problem.

- 8-1 See Figure S8-1a.
- 8-2 See Figure S8-1b.
- 8-4 $\phi = 4.9^\circ$.
- 8-6 $\phi = 13.8^\circ$.

CHAPTER 9 GEAR TRAINS

- 9-1 Pitch diameter = 4.8, circular pitch = 0.628, addendum = 0.20, dedendum = 0.25, tooth thickness = 0.314, and clearance = 0.050.
- 9-5 a. $p_d = 10$, b. $p_d = 6$
- 9-6 Assume a minimum no. of teeth = 16, then: pinion = 16t and 1.600-in pitch dia. Gear = 112t and 11.240-in pitch dia. Contact ratio = 1.68.

* These files can be found in the PROBLEM SOLUTIONS folder downloadable with this text.

**FIGURE S8-1**

Solutions to Problems 8-1 and 8-2

- 9-7** Assume a minimum no. of teeth = 16, then: pinion = 16t and 3.20-in pitch dia. Gear = 96t and 19.20-in pitch dia. An idler gear of any dia. is needed to get the positive ratio. Contact ratio = 1.67.
- 9-10** Three stages of 4:1, 4:1, and 5:1 give -80:1. Stage 1 = 20t ($d = 1.67$ in) to 80t ($d = 6.67$ in). Stage 2 = 20t ($d = 1.67$ in) to 80t ($d = 6.67$ in). Stage 3 = 18t ($d = 1.5$ in) to 90t ($d = 7.5$ in).
- 9-12** The square root of 120 is > 10 so will need three stages. $5 \times 4 \times 6 = 150$. Using a minimum no. of teeth = 18 gives 18:90, 18:72, and 18:108 teeth. Pitch dias. are 3.6, 18 and 21.6 in. An idler (18t) is needed to make the overall ratio positive.
- 9-14** The factors $4 \times 7 = 28$. The ratios 24:96 and 15:105 revert to same center distance of 7.5 in. Pitch dias. are 1.875, 3, 12, and 13.125 in.
- 9-16** The factors $6.5 \times 10 = 65$. The ratios 22:143 and 15:150 revert to same center distance of 10.3125 in. Pitch dias. are 2.75, 17.875, 1.875, and 18.75.
- 9-19** The factors $2 \times 1.5 = 3$. The ratios 15:30 and 18:27 revert to the same center distance of 3.75. Pitch dias. are 2.5, 5, 3, and 4.5. The reverse train uses the same 1:2 first stage as the forward train, so it needs a second stage of 1:2.25 which is obtained with a 12:27 gearset. The center distance of the 12:27 reverse stage is 3.25 which is less than that of the forward stage. This allows the reverse gears to engage through an idler of any suitable diameter to reverse output direction.
- 9-21** For the low speed of 6:1, the factors $2.333 \times 2.571 = 6$. The ratios 15:35 and 14:36 revert to the same center distance of 3.125. Pitch dias. are 1.875, 4.375, 1.75, and 4.5. The second speed train uses the same 1:2.333 first stage as the low-speed train, so it needs a second stage of 1:1.5 which is obtained with a 20:30 gearset which reverts to the same center distance of 3.125. The additional pitch dias. are 2.5 and 3.75. The reverse train also uses the same 1:2.333 first stage as both forward trains, so it needs a second stage of 1:1.714 which is obtained with a 14:24 gearset. The center distance of the 14:24 reverse stage is 2.375 which is less than that of the forward stages. This allows the reverse gears to engage through an idler of any suitable diameter to reverse output direction.

TABLE S9-1 Solution to Problem 9-29

Possible Ratios for Two-Stage Compound Gear Train to Give the Ratio 2.718 28

Pinion 1	Gear 1	Ratio 1	Pinion 2	Gear 2	Ratio 2	Train Ratio	Abs Error
25	67	2.68	70	71	1.014	2.718 285 71	5.71E-06
29	57	1.966	47	65	1.383	2.718 268 53	1.15E-05
30	32	1.067	31	79	2.548	2.718 279 57	4.30E-07
30	64	2.133	62	79	1.274	2.718 279 57	4.30E-07
31	48	1.548	45	79	1.756	2.718 279 57	4.30E-07
31	64	2.065	60	79	1.317	2.718 279 57	4.30E-07
31	79	2.548	75	80	1.067	2.718 279 57	4.30E-07
35	67	1.914	50	71	1.420	2.718 285 71	5.71E-06

9-25 a. $\omega_2 = 790$, c. $\omega_{arm} = -4.544$, e. $\omega_6 = -61.98$

9-26 a. $\omega_2 = -59$, c. $\omega_{arm} = 61.54$, e. $\omega_6 = -63.33$

9-27 a. 577.7 rpm and 4.33 to 1, b. $x = 577.7 \times 2 - 800 = 355.4$ rpm

9-29 See Table S9-1 for solution. The third row has the smallest error and smallest gears.

9-35 $\eta = 0.963$.

9-37 $\eta = 0.996$.

9-39 $\omega_1 = 979.6$ rpm, $\omega_2 = 2742.9$ rpm.

9-41 $\omega_1 = -293.9$ rpm, $\omega_3 = -587.8$ rpm.

9-43 $\omega_G = -18.6$ rpm, $\omega_F = -187.7$ rpm.

9-67 $\phi = 26.23^\circ$.

9-69 Gear ratio = 2.4 and contact ratio = 1.698. Circular pitch = 0.785, base pitch = 0.738, pitch dia. = 6.25 and 15, outside dia. = 6.75 and 15.5, center dist. = 10.625, addendum = 0.250, dedendum = 0.313, whole depth = 0.562 5, clearance = 0.063 (all in inches).

9-71 Four stages with factors $6 \times 5 \times 5 \times 5 \times 5 = 750$: Stage 1 = 14t to 84t. Stages 2, 3, 4 = 14t to 70t. Output in same direction as input due to even number of stages.

CHAPTER 10 DYNAMICS FUNDAMENTALS

10-1 CG @ 8.77 in from handle end, $I_{zz} = 0.394$ in-lb-sec², $k = 9.35$ in.

10-2 CG @ 8.08 in from handle end, $I_{zz} = 0.221$ in-lb-sec², $k = 8.95$ in.

10-4

a. $x = 3.547$, $y = 4.8835$, $z = 1.4308$, $w = -1.3341$

b. $x = -62.029$, $y = 0.2353$, $z = 17.897$, $w = 24.397$

10-6

- a. In series: $k_{eff} = 3.09$, Softer spring dominates
 b. In parallel: $k_{eff} = 37.4$, Stiffer spring dominates

10-9

- a. In series: $c_{eff} = 1.09$, Softer damper dominates
 b. In parallel: $c_{eff} = 13.7$, Stiffer damper dominates

10-12 $k_{eff} = 12 \text{ N/mm}$, $m_{eff} = 0.688 \text{ kg}$

10-14 $k_{eff} = 225 \text{ N/mm}$, $m_{eff} = 58.5 \text{ kg}$

10-20 Effective mass in 1st gear = 0.054 bl, 2nd gear = 0.096 bl, 3rd gear = 0.216 bl, 4th gear = 0.863 bl.

10-21 Effective spring constant at follower = 308.35 lb/in.

10-25 Effective spring constant = 111.1 N/mm, effective mass = 27 kg.

10-26 $x = 5.775 \text{ in}$.

10-34 I_{crank} about pivot = 1 652 kg-mm², I_{rocker} about pivot = 18 420 kg-mm², $I_{coupler}$ about CG = 2106 kg-mm² (both couplers are the same).

10-35 $x = 774 \text{ mm}$ to strike point of ball.

CHAPTER 11 DYNAMIC FORCE ANALYSIS

11-3 Open file P11-03row.sld in program LINKAGES to check your solution.*

11-4 Open file P11-03row.sld in program LINKAGES to check your solution.*

11-5 Open file P11-05row.4br in program LINKAGES to check your solution.*

11-6 Open file P11-05row.4br in program LINKAGES to check your solution.*

11-7 Open file P11-07row.4br in program LINKAGES to check your solution.*

11-12 $F_{12x} = -1851 \text{ N}$, $F_{12y} = 1315 \text{ N}$; $F_{14x} = 1047 \text{ N}$, $F_{14y} = -3156 \text{ N}$;
 $F_{32x} = 479 \text{ N}$, $F_{32y} = -275 \text{ N}$; $F_{43x} = 53.7 \text{ N}$, $F_{43y} = -1087 \text{ N}$; $T_{12} = -45.3 \text{ N-m}$

11-13 Open file P11-13.4br in program LINKAGES to check your solution.*

11-14 $F_{12} = 1\,308 \text{ lb}$, $F_{32} = 1\,290 \text{ lb}$, $F_{43} = 1\,290 \text{ lb}$, $F_{14} = 710 \text{ lb}$,
 $F_{hand} = 63.2 \text{ lb}$, $JFI = 0.645$.

11-25 $T_{12} = 463 \text{ lb-in}$

11-40 Mass moment of inertia needed in flywheel = 11.8 bl-in². Many flywheel geometries are possible. Assuming a steel cylinder with a radius of 9.0 in, thickness = 1.474 in.

CHAPTER 12 BALANCING**12-1**

- a. $m_b r_b = 0.934$, $\theta_b = -75.5^\circ$
 c. $m_b r_b = 5.932$, $\theta_b = 152.3^\circ$
 e. $m_b r_b = 7.448$, $\theta_b = -80.76^\circ$

* These files can be found in the PROBLEM SOLUTIONS folder downloadable with this text.

12-5

- a. $m_a r_a = 0.814$, $\theta_a = -175.2^\circ$, $m_b r_b = 5.50$, $\theta_b = 152.1^\circ$
 c. $m_a r_a = 7.482$, $\theta_a = -154.4^\circ$, $m_b r_b = 7.993$, $\theta_b = 176.3^\circ$
 e. $m_a r_a = 6.254$, $\theta_a = -84.5^\circ$, $m_b r_b = 3.671$, $\theta_b = -73.9^\circ$

12-6 $W_a = 3.56$ lb, $\theta_a = 44.44^\circ$, $W_b = 2.13$ lb, $\theta_b = -129.4^\circ$

12-7 $W_a = 4.2$ lb, $\theta_a = -61.8^\circ$, $W_b = 3.11$ lb, $\theta_b = 135^\circ$

12-8 These are the same linkages as in Problem 11-5. Open the file P11-05row.4br in program LINKAGES to check your solution.* Then use the program to calculate the flywheel data.

12-9 Open the file P12-09.4br in program LINKAGES to check your solution.*

12-14 $R_3 = 5.85$ in, $\theta_3 = -142.11^\circ$, $R_4 = 1.13$ in, $\theta_4 = 120^\circ$

12-16 $W_4 = 14.48$ lb, $\theta_4 = 89.15^\circ$, $W_5 = 5.04$ lb, $\theta_5 = 83.90^\circ$

12-18 $d_3 = 18.95$ mm, $\theta_3 = -147.46^\circ$, $d_4 = 20.8$ mm, $\theta_4 = 28.94^\circ$

12-38 Plane 2: $e = 0.113$, $\theta = -152.15^\circ$. Plane 3: $e = 0.184$, $\theta = 19.36^\circ$.

CHAPTER 13 ENGINE DYNAMICS

13-1 Exact solution = $-42\,679.272$ in/sec @ 299.156° and 200 rad/sec

Fourier series approximation = $-42\,703.631$ in/sec @ 299.156° and 200 rad/sec

Error = -0.0571% ($-0.000\,571$)

13-3 Gas torque = 2040 (approx.), Gas force = 3142

13-5 Gas torque = 2039.53 (approx.), Gas torque = 2039.91 (exact)

Error = 0.0186% ($0.000\,186$)

13-7

- a. $m_b = 0.007\,48$ at $l_b = 7.2$, $m_p = 0.012\,51$ at $l_p = 4.31$
 b. $m_b = 0.008\,00$ at $l_b = 7.2$, $m_a = 0.012\,00$ at $l_a = 4.80$
 c. $I_{model} = 0.691\,2$, Error = 11.48% ($0.114\,8$)

13-9 $m_{2a} = 0.018$ at $r_a = 3.5$, $I_{model} = 0.220\,5$, Error = -26.5% (-0.265)

13-11 Open the file P13-11.eng in program ENGINE to check your solution.*

13-14 Open the file P13-14.eng in program ENGINE to check your solution.*

13-19 Open the file P13-19.eng in program ENGINE to check your solution.*

* These files can be found in the PROBLEM SOLUTIONS folder downloadable with this text.

TABLE S15-1
Solutions to Problem
15-6

	ω_n	ω_d	c_c
<i>a</i>	3.42	3.38	8.2
<i>b</i>	4.68	4.65	19.7
<i>c</i>	0.26	0.26	15.5
<i>d</i>	2.36	2.33	21.2
<i>e</i>	5.18	5.02	29.0
<i>f</i>	2.04	1.96	49.0

CHAPTER 14 MULTICYLINDER ENGINES

Note: Use program ENGINE to check your solutions.

14-23 *mr product* on the balance shafts = $5.017\text{E-}3$ bl-in or 1.937 lb-in.

CHAPTER 15 CAM DYNAMICS

15-1 to **15-5** Use program DYNACAM to solve these problems. There is not any *one right answer* to these design problems.

15-6 See Table S15-1.

15-7 to **15-19** Use program DYNACAM to solve these problems. There is not any *one right answer* to these design problems.

Appendix G

EQUATIONS FOR UNDER-OR OVERBALANCED MULTICYLINDER ENGINES

G.1 INTRODUCTION

Chapter 14 developed the equations for shaking forces, moments, and torques in multicylinder engines of inline and vee configurations. In Chapter 14, it is assumed that the crank throws are all exactly balanced, an assumption that greatly simplifies the equations. However, some multicylinder engines overbalance the crank throws to reduce main bearing forces. This also can have an effect on shaking forces and moments.

This appendix provides replacement equations for the simplified versions in Chapter 14, and these equations do not assume exactly balanced crank throws.* The equation numbers used here correspond to those in Chapter 14 and can be substituted for the simplified ones if desired. In the equations that follow, m_A is the effective crank pin mass and m_B the effective wrist pin mass as defined in Chapter 13. The parameters m_C and r_C represent, respectively, the counterweight mass of any one crank throw and the radius to the counterweight's *CG*. All other parameters are the same as defined in Chapters 13 and 14.

* These complete equations are used in program LINKAGES.

For an inline engine (Section 14.3) the shaking forces for an engine with an under- or overbalanced crankshaft are:

$$\begin{aligned}
 F_{s_x} \cong & (m_A + m_B) r \omega^2 \left[\cos \omega t \sum_{i=1}^n \cos \phi_i + \sin \omega t \sum_{i=1}^n \sin \phi_i \right] \\
 & + m_c r_c \omega^2 \left[\cos(\omega t + \pi) \sum_{i=1}^n \cos \phi_i + \sin(\omega t + \pi) \sum_{i=1}^n \sin \phi_i \right] \\
 & + \frac{m_B r^2 \omega^2}{l} \left[\cos 2\omega t \sum_{i=1}^n \cos 2\phi_i + \sin 2\omega t \sum_{i=1}^n \sin 2\phi_i \right] \hat{\mathbf{i}}
 \end{aligned} \tag{14.2d}$$

$$\begin{aligned}
 F_{s_y} \cong & m_A r \omega^2 \left[\sin \omega t \sum_{i=1}^n \cos \phi_i - \cos \omega t \sum_{i=1}^n \sin \phi_i \right] \\
 & + m_c r_c \omega^2 \left[\sin(\omega t + \pi) \sum_{i=1}^n \cos \phi_i - \cos(\omega t + \pi) \sum_{i=1}^n \sin \phi_i \right] \hat{\mathbf{j}}
 \end{aligned}$$

For an inline engine (Section 14.3) the shaking moments for an engine with an under- or overbalanced crankshaft are:

$$\begin{aligned}
 M_{s_x} \cong & (m_A + m_B) r \omega^2 \left[\cos \omega t \sum_{i=1}^n z_i \cos \phi_i + \sin \omega t \sum_{i=1}^n z_i \sin \phi_i \right] \\
 & + m_c r_c \omega^2 \left[\cos(\omega t + \pi) \sum_{i=1}^n z_i \cos \phi_i + \sin(\omega t + \pi) \sum_{i=1}^n z_i \sin \phi_i \right] \\
 & + \frac{m_B r^2 \omega^2}{l} \left[\cos 2\omega t \sum_{i=1}^n z_i \cos 2\phi_i + \sin 2\omega t \sum_{i=1}^n z_i \sin 2\phi_i \right] \hat{\mathbf{i}}
 \end{aligned} \tag{14.6b}$$

$$\begin{aligned}
 M_{s_y} \cong & m_A r \omega^2 \left[\sin \omega t \sum_{i=1}^n z_i \cos \phi_i - \cos \omega t \sum_{i=1}^n z_i \sin \phi_i \right] \\
 & + m_c r_c \omega^2 \left[\sin(\omega t + \pi) \sum_{i=1}^n z_i \cos \phi_i - \cos(\omega t + \pi) \sum_{i=1}^n z_i \sin \phi_i \right] \hat{\mathbf{j}}
 \end{aligned}$$

For a vee or opposed engine (Sections 14.7 and 14.8) the shaking forces for an engine with an under- or overbalanced crankshaft are:

$$F_{s_x} \equiv (F_{s_L} + F_{s_R}) \cos \gamma + m_A r \omega^2 \left[\cos \omega t \sum_{i=1}^n \cos \phi_i + \sin \omega t \sum_{i=1}^n \sin \phi_i \right] \\ + m_c r_c \omega^2 \left[\cos(\omega t + \pi) \sum_{i=1}^n \cos \phi_i + \sin(\omega t + \pi) \sum_{i=1}^n \sin \phi_i \right] \hat{\mathbf{i}} \quad (14.10j)$$

$$F_{s_y} \equiv (F_{s_L} - F_{s_R}) \sin \gamma + m_A r \omega^2 \left[\sin \omega t \sum_{i=1}^n \cos \phi_i - \cos \omega t \sum_{i=1}^n \sin \phi_i \right] \\ + m_c r_c \omega^2 \left[\sin(\omega t + \pi) \sum_{i=1}^n \cos \phi_i - \cos(\omega t + \pi) \sum_{i=1}^n \sin \phi_i \right] \hat{\mathbf{j}} \\ \mathbf{F}_s = F_{s_x} \hat{\mathbf{i}} + F_{s_y} \hat{\mathbf{j}}$$

For a vee or opposed engine (Sections 14.7 and 14.8) the shaking moments for an engine with an under- or overbalanced crankshaft are:

$$M_{s_x} \equiv (M_{s_L} + M_{s_R}) \cos \gamma + m_A r \omega^2 \left[\cos \omega t \sum_{i=1}^n z_i \cos \phi_i + \sin \omega t \sum_{i=1}^n z_i \sin \phi_i \right] \\ + m_c r_c \omega^2 \left[\cos(\omega t + \pi) \sum_{i=1}^n z_i \cos \phi_i + \sin(\omega t + \pi) \sum_{i=1}^n z_i \sin \phi_i \right] \hat{\mathbf{i}} \quad (14.11c)$$

$$M_{s_y} \equiv (M_{s_L} - M_{s_R}) \sin \gamma + m_A r \omega^2 \left[\sin \omega t \sum_{i=1}^n z_i \cos \phi_i - \cos \omega t \sum_{i=1}^n z_i \sin \phi_i \right] \\ + m_c r_c \omega^2 \left[\sin(\omega t + \pi) \sum_{i=1}^n z_i \cos \phi_i - \cos(\omega t + \pi) \sum_{i=1}^n z_i \sin \phi_i \right] \hat{\mathbf{j}} \\ \mathbf{M}_s = M_{s_x} \hat{\mathbf{i}} + M_{s_y} \hat{\mathbf{j}}$$

Note that inertia torque is unaffected by crankshaft balance condition because, at constant angular velocity, the acceleration vector of the crank pin mass is centripetal and has no moment arm. The moment of inertia added to the crankshaft by any overbalance mass will increase the flywheel effect of the crankshaft and thus reduce its willingness to change rotational speed in transient angular acceleration. But, the size of the engine's physical flywheel can be reduced to compensate for the more massive crankshaft.

INDEX

A

acceleration 3, 178, 357

absolute 359, 364

analysis

analytical 365

graphical 360, 361

angular 590, 599, 608

as free vector 365

definition 357

inverted slider-crank 376, 378

cam-follower 416

comparison of shapes 434

peak factor 432

centripetal 358, 364, 579

coriolis 373.

See Also coriolis acceleration

difference

analytical solution 364, 366

definition 360

equation 359

graphical solution 364

in slider-crank 370

discontinuities in 422

human tolerance of 382

linear 590, 599

modified sine 427, 434

modified trapezoidal 427

normal 364

of any point on link 380

of a valve train 785

of geared fivebar 379

of piston 384

of slip 378

relative 359, 364

sinusoidal 425

tangential 358, 364

tolerance 383

accelerometer 382

across variable 568

actuator 38, 582

addendum 493, 497

circle 497

modification coefficients 502

AGMA 497, 502

air

cylinder 38, 100

motor 79

all wheel drive 537

all-wheel-drive 537

Ampere, Andre Marie 5

amplitude ratio 772

analogies 11, 763

analog 569

analysis 11, 12, 24, 98, 99, 589

definition 8

of mechanisms 3, 30

analytical linkage synthesis 100, 102, 233, 236, 245.

See Also linkage: synthesis

compare to graphical 243

angle

of approach 494

of a vector

definition 189

of recess 494

angular velocity ratio 309, 310, 311, 492

definition 307

antiparallelogram 57

linkage 315

apparent position 183

applications

assembly machines 452

automobile engine 409

automobile suspension 128, 311

automobile transmission 493

engine valves 443

indexing table drive 435

movie camera 126

of air motors 79

of fluid power cylinders 79

of hydraulic motors 79

of kinematics 6

of solenoids 80

optical adjusting mechanism 312

steam locomotives 57

toggle linkages 103

approximate

circle arc 149, 151

dwell 152

straight line 125, 141, 144, 151

arc of action 494

arctangent

two-argument code for 180

arm (epicyclic) 521.

See Also gear: train: epicyclic

Artobolevsky 6

linkage catalog 153
reference 158

asperities 33

atlas of coupler curves
fourbar 126
geared fivebar 131

automobile
clutch 622
suspension 128, 141, 311
wheel balancer 661

axis
of rotation 558
of slip
 cam-follower 304, 461
 inverted slider-crank 330, 374, 610
 slider block 318, 319
of transmission 495
 cam-follower 461, 462
 gear teeth 494
 inverted slider-crank 330, 610
 slider block 318, 319

axle 311

B

babbitt 711

backdrive 102, 506

background research 9, 16, 22, 639

backhoe 7

backlash 497, 498.
 See Also gear: antibacklash
 definition 497

balance
complete 642
dynamic 642, 646
 tires 662, 663
mass 643
shafts 736, 752, 755
single-plane 643
static 642, 643, 755
 tires 661

balancer
Lanchester 756
Nakamura 756

balancing 578
dynamic 649, 755
 secondary force 756
engines
 multi-cylinder 751
 single-cylinder 705

linkages 651
 effect on input torque 656, 657
 effect on pin forces 655
 optimum counterweights 657
 shaking force 651
 shaking moment 657
static 645

ball
and socket 33
joint 33. *See Also* joint

bank
angle 742. *See Also* vee: angle
engine 721. *See Also* engines: vee
of cylinders 674

Barker 60

base circle
cam 460, 473
gear 497, 500, 507
involute 493
radius 460

base units 17

BDC. See bottom dead center (BDC)

beam
cantilever 574
double cantilever 574
indeterminate 42
simply supported 42

bearing 69
ball 69
bushing 72
effective diameter 72
effective length 72
flange-mount 69
journal 69
linear ball 70
pillow block 69
ratio 72, 73
 definition 72
 poor, example of 73
roller 69
rolling-element 69
sleeve 69
spherical rod end 69

belt 6, 32, 490
flat 509
synchronous 509
timing 509
vee 491, 509
vibration in 511

benchmarking 9

Berkof-Lowen method 651

BFI. See Brute Force and Ignorance

big end (conrod) 689

binary 42
link 32

binomial
expansion 683
theorem 682, 687

bisector 151

blank paper syndrome 8, 624

blobs 18

bore/stroke ratio 709

Boston rocker 316

bottom dead center (BDC) 675

boundary conditions 409, 416, 439, 445

brainstorming 11

brake 521

branch
defect 212
definition 212

building blocks 36

bump steer 311

Burmester curves 266

bushing 72
ball 145

C

CAD. See Computer Aided: Drafting

CAE. See Computer Aided: Engineering

cam 6, 32, 100, 148, 385, 409, 763
and follower 100, 148, 304
automotive valve 574, 776
axial 414, 476
barrel 414, 435
conjugate 476
contour 473
cylindrical 414
definition 409
design
 fundamental law of 420
desmodromic 780
disk 565, 764, 775
double-dwell 417
face 414
force-closed 780, 783
form-closed 781, 783
ground 477
mechanisms 568

- milled 477
- motion program types 415
- plate 414, 783
- radial 414, 476
- single-dwell 443
 - asymmetrical rise-fall* 448
 - symmetrical rise-fall* 443
- stationary 470
- track 780
- CAM. *See* Computer Aided: Manufacturing**
- cam-follower** 52, 53, 55, 73
- camshaft** 780
- capacitor** 569, 571
- carburetor** 677
- carry through** 57
- cartesian**
 - coordinates 188
 - form 180
- Cayley diagram** 134, 137.
 - See Also* cognate
- degenerate 137
- center**
 - of curvature 472
 - of gravity 556, 599, 687
 - global* 557
 - of percussion 563, 659, 689
 - of rotation 563, 564, 659, 689
 - point 263
 - circle* 263
- centrifugal force** 578, 752
- centrodes** 313. *See Also* polodes
 - fixed, moving 315
 - noncircular gears 508
- centros** 298.
 - See Also* instant centers
- CEP cams. *See* Critical Extreme Position (CEP)**
- chain** 32
 - drive 490, 510
 - vibration in* 511
 - silent 510
- change points** 57
- characteristic equation** 768, 769
- Chasles**
 - reference 218
 - theorem 185, 297
- Chebyshev** 134, 140, 144, 263.
 - See Also* Roberts: -Chebyshev theorem
- chordal action** 510
- circle**
 - arc 144
 - with remote center* 125
 - point 263
 - circle* 263
- circuit**
 - defect 212
 - definition 212
 - of a linkage 193
 - distinguishing* 215
 - number of* 213
- circular**
 - gears 316
 - pitch 498
- civil engineering** 5
- clearance** 497
- Clerk** 677
 - cycle 675
 - engine 677
- clockworks** 5
- closed**
 - curve 125
 - kinematic chains 38
 - mechanism 38
- closed loop** 78
- clutch**
 - automobile 622, 699
 - synchromesh 532, 533
- CNC. *See* Continuous Numerical Control (CNC)**
- coefficient**
 - of damping 568, 573. *See Also* resonance
 - of fluctuation 621, 622, 697. *See Also* flywheel
- cognate** 134, 137.
 - See Also* Cayley diagram;
 - See Also* Roberts: -Chebyshev theorem;
 - See Also* Roberts: diagram
- character of 134
- fourbar linkage 134
- geared fivebar 140
- colinearity** 102, 119
- combined functions**
 - for cams 425
- common**
 - normal 304, 494, 495
 - tangent 304, 493
- communication** 17, 24
- complex**
 - motion 31
 - conrod* 688
 - coupler* 36, 124
 - definition* 32, 184
 - number 188
 - notation* 188, 189
 - plane 189
- compliance**
 - definition 65
- compliant mechanisms** 65
- component**
 - orthogonal 196
- compound**
 - epicyclic train 525
 - gear train 512. *See Also* gear: train
- compression**
 - ignition 679
 - stroke 676
- Computer Aided**
 - Drafting 99, 624
 - Engineering 99, 100, 126
- computer graphics** 119
- computer programs. *See* programs**
- concave** 466
- conjugate**
 - action 506
 - cams 413, 780
- conjugates** 413, 493
- connecting rod** 36, 674
- conrod** 679, 688, 691
 - two per crank throw 746
- conrod/crank ratio** 709, 710
- conservation of energy** 580
- conservative model** 764
- constant**
 - acceleration 426
 - of integration 766
 - velocity 452
- constrained** 38
- construction angle** 120
- contact ratio** 502
 - minimum 502
- continuation methods** 272
- continuous** 53
 - motion 452, 453
- convex** 466
- coordinate system** 32, 180, 590
 - absolute 17, 180
 - global 180, 237, 605
 - local 180

nonrotating 180, 590, 605
rotating 180, 605

coriolis acceleration 373, 375

correction planes 647, 755

cost 100

coulomb friction 566, 606.

See Also friction: non-linear

counter

balance 65
couple 647
rotating eccentrics 752
shaft 532
weight 644, 656, 708, 755
crankshaft 706
optimum balance 657

coupler 22, 124, 126, 207

as a physical pendulum 658
attachment points
 alternate 114
curve equation 273
curves 125, 126, 128, 131,
 134, 151
 atlas of 126
 degenerate 124
 degree of 124
 design charts 131
 symmetrical 129
definition 36
output 105, 107, 111, 112
point 126, 151

coupler curve

equation
 complexity 273
synthesis 274

CPM cams.

See Critical Path Motion

crank 126, 674, 688

definition 36
eccentric 71
phase diagram 723, 738
short 71
throw 719

crank-conrod ratio 682

crankpin 691

splayed 755

crank-rocker 57, 126

crankshaft 677, 719

balance weights 705
mirror symmetric 738, 747
phase diagram 722

crank-shaper 55

crank-slider.

See Also slider-crank

constant velocity slider 153
quick-return 122
threebar 593

creative process 11, 21

definition 21

creativity 7, 10, 11, 21, 28

Critical Extreme Position (CEP)

410, 414, 417

critically damped.

See damping: critical

Critical Path Motion (CPM)

410, 414, 452, 453, 458

crossed

helical gears 505
mechanism 193, 256

crossover shock 476, 497, 782

crowned pulley 509

crunode 125, 126, 131

cubic function

finding roots of 214

current 568

curvilinear translation 137, 140

cuspid

on cam 467
on coupler curve 128, 131
on moving centrode 316

cycloid

curve 125
gear tooth 493

cycloidal

coupler curve 316
displacement 422, 425, 443
 compared 434
 dynamic torque 784
 single-dwell 442

cylinder

air and hydraulic 79.
See Also air: cylinder;
See Also hydraulic: cylinder

D

d'Alembert 5, 578, 579, 581, 64
3, 691

dampers 574, 772

combining 569
in parallel 570
in series 569

damping 566, 568, 574, 763, 767,
775

coefficient 568
critical 768, 769
effective 570, 573
internal 576
nonlinear 566
pseudo-viscous 568
quadratic 566
ratio 768, 772, 774, 775
viscous 566

DC component 683

decision matrix 13

dedendum 497, 500

circle 498

deferred judgment 11

deflection 555, 566

bending 574
torsional 576

degree 124, 131, 439

degree of freedom 30, 35, 40

definition 37
distribution of 48
spatial mechanisms 40
visualizing 35

DeJonge 6

Delone 156

delta

phase angle 722
 optimum 722
power stroke angle 732
triplet 46, 48, 52

deltoid 59

Denavit, J. 6, 134

density

mass 555
weight 555

derived unit 17

descriptive geometry 5, 24

design 7, 21, 100

axiomatic 15
by successive analysis 99, 100
case study 21
computer-aided 13
definition 7
detailed 13, 15, 26
process 3, 7, 8, 14, 28, 99, 555
qualitative 99
ratios 709
simplicity in 56
specifications 10
trade-off 73, 621, 709

desmodromic 413, 476, 780.

See Also cam: form-closed

determinant 259

diagrams

kinematic
drawing 36

diametral pitch 498

Diesel cycle 679

differential 521.

See Also gear: train: epicyclic

automotive 537
center 537
rear 537
center 537
definition 536
limited slip 538
Torsen 538

dimensional synthesis 100, 102,
104, 158. *See Also* analytical:
linkage synthesis

of a fourbar linkage 104

Dirac delta functions 420

discontinuities 420

discriminant 769

displacement

cam 416, 438
definition 181
total 185

dissipative element 772

Dixon, A. 15

DOF. *See* degree of freedom

dot product 580

double

crank 57.
dwell 153, 417
cam 417
linkage 151, 416
enveloping wormset 506
harmonic 444, 784
parallelogram linkage 59
rocker 57, 103, 109, 141.

See Also crank-rocker

dragged crank 121

drag link 57, 121

driver 74

crank 601
stage 109

driving torque 606, 686.

See Also torque: driving

duplicate planar linkages 101

dwell 53, 73, 148, 151, 415, 477

cam

double- 417
single 443
single- 443

definition 148

linkages 148
mechanism 53, 148, 152
motion 125

dyad 105, 238, 240, 243, 251, 263

definition 38
driver 109, 111, 114
analytical synthesis 234
output 121, 122, 148

Dynacam program

example
constant velocity 454
force 776, 777, 780
polynomial 442
radius of curvature 469
single-dwell 442
torque 783, 785
general information 817, 818

dynamic

analysis 26
balance. *See* balance: dynamic
balancing. *See* balancing: dynamic
devices requiring 647
machine 662
equilibrium 578
equivalence 659
requirements for 688
force 3, 4, 18, 178, 382, 384, 555
analysis 553, 589
measurement 785
friction 566
models 554
system 4, 578, 590

E

eccentric

crank 71
masses 752, 753

eccentricity

cam-follower
definition 461
effect on pressure angle 464
flat-faced 465, 474
roller 461

effective

damping 573
linkage 409, 491
links 52, 307

mass 571

spring 573

efficiency 505, 580

definition 529
of a conventional gear train 529
of an epicyclic train 529

elastomers 509

electrical circuit 569

electric motors 74, 617.

See Also motor

electromechanical devices 101

encoder 795

endpoint specification 414

energy

kinetic 291
in cam-followers 426, 435, 459
in flywheels 618, 697
in lever ratios 571
in resonance 772
in rotating systems 558
in virtual work 580, 581
peak 434
law of conservation 580
method 580
potential 580, 772
storage elements 773

engineering 14

approach 14
design 4, 7, 98, 148
cost in 100
definition 7
human factors 16
report 17

Engine program 672, 683, 693,
717

flywheel calculations 697

engines 74, 674

inline 719
four-cylinder 756
six-cylinder 752
multicylinder 719
balancing 751
opposed 721
twin 751
radial 721
rotary 721
vee 674, 721, 739
eight 674, 739, 746, 747, 754
, 755
six 739, 755
twelve 752
twin 754

epicyclic gear train.

See gear: train: epicyclic
efficiency of 530

equation solver 217, 577, 624

equilibrium 566, 578

E-quintet 46

equivalent

mass 571
spring 571
system 568, 573, 763

Erdman, Arthur G. 6, 119,
233, 266

ergonomics 16. *See Also* human
factors engineering

euclidean geometry 105

Euler 5

equivalents 191
identity 189
theorem 185

Eureka! 11, 23

Evans, Oliver 5

straight-line linkages 144

even firing. *See* firing: even

evolute 493

exact straight line. *See* straight-line:
mechanisms: exact

Example 16-1 800

Example 16-2 803

Example 16-3 807

exhaust stroke 677

external

gearset 499
load 599
torque 599

F

face width (gears) 498

Ferguson's paradox 525, 528

film advance mechanism 126

finite

difference method 16
element method 16

firing

even 730
importance of 733
inline four 736, 739
vee eight 750
vee engines 750
order 736, 748

pattern 730

uneven 733

first moment of mass.

See mass: moment

fivebar linkage

geared 63

fixed

centrode 313, 316.

See Also centrodes

pivots 112, 114, 126, 134
specified 115.

See Also *specified fixed pivots*

flat belts. *See* belt: flat

flat-four engine.

See engines: opposed

flexure hinge 22

fluctuation 621. *See Also* coeffi-
cient: of fluctuation

flywheel 77, 559, 613

calculation

in program Dynacam 783
in program Fourbar 617

designing

for fourbar linkage 616
for IC engine 697

effect 656

engine 697, 699

in IC engines 752

materials 622

moment of inertia of 697

physical 622

sizing 621

follower 763

cam 101, 409

aligned 461

force-closed 764, 775

form-closed 780

system 409, 763

underdamped 774

flat-faced 413, 470, 476

float 476

force- or form-closed? 476

jump 476, 774, 776, 778

mushroom 413

roller 780

rotating 410

slip 780

translating 410

flat-faced 574

precession 475

roller 783

translating or rotating? 475

foot-pound-second (fps) system 17

force

analysis

kinetostatic 775, 776, 780, 781,
800, 803

applied 591, 594

centrifugal 579, 691

closed 35

closure 35, 412, 780

crankpin 701, 703, 705

dynamic 672

cam-follower 776, 777, 778,
781, 783, 785

compared to gravitational 581
minimizing 425

external 581, 643

externally applied 591

gas. *See* gas: force

gravitational 580, 590

impact 776

inertia 571, 581, 625
, 643, 685, 691, 703

link 129

mainpin 703

effect of balancing on 709

piston sidewall 701

primary 739

reaction 601

secondary 739

shaking 612, 642, 672

cancelling 752, 753

fourbar linkage 656

in inline engines 723, 739

in one-cylinder engines 685, 691

in vee engines 744

primary 736

secondary 736

spring 565, 764, 775

transducer 662, 785

transmission 104

wristpin 701, 705

forcing frequency.

See frequency: forcing

form-closure 35, 412, 780

Formula 1

engine redline 684

forward dynamic analysis.

See force: analysis: dynamic

fourbar linkage 55, 134, 599, 642

acceleration 365

anti-parallellogram 57

change points 57

classification of 60

cognates 137

character of 137

coupler curves 125
 crank-rocker 50, 102
 double-rocker 102, 103, 105
 Grashof condition 55
 linkless 313
 mechanism 22
 optimum straight-line 144
 quick-return 119, 120
 subchain 105
 symmetrical 130
 triple-rocker 57, 103

Fourbar program
 example
 three-position synthesis 256
 two-position synthesis 245

Fourier 5, 683
 descriptors 276
 series 683, 726
 equation 683

four-position synthesis.
See synthesis: four-position

four-stroke cycle 675, 733

free body 579, 593
 diagram 590, 599,
 691, 700, 764, 767

free choices 106
 for function synthesis 268
 in three-position synthesis 250, 252
 in two-position synthesis 239,
 240, 242

free vector 298, 363, 365

frequency
 forcing 772, 774
 fundamental 766, 774, 779
 natural 763
 and resonance 774
 cam-follower 775, 777, 780, 789
 circular 766, 771
 undamped 766, 767
 overtones 766
 ratio 772
 response 785

Freudenstein, F. 6, 192

friction 35, 491, 691
 belts 491
 Coulomb 566
 force 596
 in linkages 104
 nonlinear 566
 work 318

frisbee 298

frustration 10, 11, 23

full joint 33, 39.
See Also joint

function
 forcing 771, 789
 generation 100, 414.
See Also motion: generation;
See Also path: generation
analytical synthesis 267
definition 101, 233
table of free choices 268
two-position 105
 generator 100, 267, 409, 796

functional visualization 10

fundamental frequency 683.

fundamental law
 of cam design 422, 774
 of gearing 492, 494, 496
 definition 494
 of servomechanism design 797

fuzzy logic 271, 274

G

Galloway mechanism 59

gas
 force 679, 685, 693, 703, 730
 curve 679, 685
 pressure 685
 curve 677, 679, 685
 torque 685, 687, 730. *See*
 Also torque

gate 128

Gauss-Jordan elimination 246

gear
 antibacklash 497.
See Also backlash
 base pitch 502
 bevel 507, 508
 spiral 507
 straight 507
 blank 507
 helical 505
 herringbone 505
 hypoid 508
 idler 512
 rack 507
 ratio 131, 204, 571
 set 141, 315
 shaper 500
 spur 505
 teeth 493
 full-depth 500
 HPSTC 503
 unequal-addendum 500, 502
 tooth action 490
 train 490, 511
 compound 512
 design algorithm 517
 earliest known reference 490
 epicyclic 521, 523, 533
 error in center distance 496
 irrational ratio 518
 reverted 515, 532
 simple 511
 worm 506
 wormset 506

gearbox 490, 513

geared fivebar
 coupler curves 131
 mechanism 62, 204
 analysis 204, 379
 cognate of fourbar 140
 coupler curves 124
 inversions of 63

gearing
 fundamental law of 492
 definition 494

gears 32, 492
 non-circular 315, 508
 profile-shifted 502
 worm. *See worm*

gearset 492, 506
 angle of approach 494
 angle of recess 494
 arc of action 494
 changing center distance 495
 contact ratio 502, 504
 external 493
 highest point of single-tooth con-
 tact 502
 internal 493, 499
 length of action 494, 502
 pressure angle 495

genetic algorithms 271

Geneva
 mechanism 53
 wheel 53

global mass center 651.
See Also center: of gravity

goal statement 10, 639

graphical
 dimensional synthesis 100
 compare to analytical 243
 tools needed for 105, 111
 position analysis 179

Grashof 56, 178

condition 56, 59, 111
 geared fivebar 63
 crank-rocker 143
 double-rocker 210
 fivebar 141
 linkage 74
 special-case 57
gravitational
 constant 17, 18
 system 17
gravity 383
ground
 definition 36
 pivots 115
 plane 134
Gruebler 38
 criterion 46
 equation 39, 42, 43

H

Hachette 5
Hain, Kurt 6
 linkages 153
 reference 157
half
 joint 35, 43
Hall, Allen S. 6, 119, 120, 140
Hammond, T. 131
harmonic 444
 number 751
harmonics 683, 726
Hartenberg, Richard 6, 134
Hart inversor 144
helical motion 33
helix angle 33, 505, 506
higher pair 6, 33, 35. *See*
 Also joint
Hitchcock chair 316
hob (gear) 500
hodograph. *See* polar: plot
Hoeken
 linkage 143, 144, 145
 reference 157
homogeneous 766
 ODE 766
 solution 767
homotopy methods 272
hood hinge 101
hood hinge mechanism 65

Hrones
 reference 157
Hrones and Nelson atlas 126
human factors engineering 16, 29
Humvee 538
hunting 497
hydraulic
 cylinder 100. *See Also* cylinder: air
 and hydraulic
 motor 74, 79. *See Also* motors: air
 and hydraulic
hyperboloids 508
hypoid gears 508

I

idea generation 11
ideation 10, 11, 15, 23
 and invention 10
identification of need 8
identity matrix 246
idler gear. *See* gear: idler
imaginary axis 189
imbalance 661
inch-pound-second (ips) 18
inclined plane 5
incubation 12, 23
indeterminate beam.
 See beam: indeterminate
indexers 435
indexing 148
 table 435
indices of merit 311
induction system 676
inductor 569, 570
inertia
 balance 736
 vee engine 739, 750
 force 578, 643, 693, 721, 730.
 See Also force: inertia
 mass moment of 558
 torque 578, 581, 694, 727.
 See Also torque: inertia
inertial reference frame 180
infinity of solutions 242, 263
inflection points 446, 469
initial conditions 766
inner ear 382

input torque 616, 656.
 See Also torque: input
instant centers 298, 299
 cam-follower 304, 463
 fourbar linkage 22, 299, 313
 generate centrodes 315
 permanent 299, 301
 slider-crank 301
 using in linkage design 311
intake stroke 676
interference 500
intermittent motion 53, 452
internal combustion engine 606,
 672, 717.
 See Also engines
internal gearset.
 See gearset: internal
invention 7, 10
inverse dynamics 554, 589, 775.
 See Also force: analysis:
 kinetostatic
inversion
 definition 53
 for three-position synthesis 114,
 115, 119
 in ideation 11
 of slider-crank 122
 force analysis 608
 position solution 202
inversions
 distinct 55
 of fourbar linkage 57
 of sixbar linkages 55
inverted slider crank.
 See slider-crank: inverted
involute 493, 496, 500, 507
 definition 493
 teeth 495
isomer 47
 invalid 48
 number of valid isomers 48
iteration 8, 12, 13, 99, 104
 , 106, 453, 555, 624

J

Jacobian 216, 217
jerk 385, 442
 angular 385, 386
 cam 416
 difference 387

in belts and chains 511
linear 386

jitter 151

joint 6, 33. *See Also* pairs

cam-follower 70
cantilevered 71
force-closed 35
force index 208, 623
form-closed 35
multiple 40, 43
one-freedom 33
order 35
slider 33
sliding 301
straddle mounted 71
two-freedom 33

joystick 33

K

Kant 5

Kaufman, R. 119, 233

Kempe 124

reference 157

Kennedy, Alexander 6

Kennedy's rule 299, 301

kinematic

applications 6
chain 35
 class of 56, 63, 64
 definition 35
 inversion of 57
pair 6, 33. *See Also* pairs
structure 40, 42
synthesis 589. *See Also* synthesis

kinematics 3, 4, 5, 553, 763,

776, 785

definition 3
diagrams
 drawing 36
history of 5, 27

kinetic energy. *See* energy: kinetic

kinetics 3, 5, 553

kinetostatics 554, 589, 775, 783.

See Also force: analysis:

kinetostatic;

See Also inverse dynamics

Kinsyn 119

KISS 15

Koster 568, 775, 789

Kota, S. 131

Kutzbach 40

L

Lagrange 5

Lanchester

engine 758
Frederick 535, 758
harmonic balancer 756, 759

L'Ecole Polytechnic 5

length of action 494, 502

Levai

12 basic epicyclic trains 521

lever 5

ratio 571, 575

limit stops 128

Lincages 119, 266

line

contact 33
of action 495
of centers 126

linear

acceleration 357, 383
actuator 79
ball bearings 70
Geneva mechanism 53
graph 299
jerk 385
motion 79
velocity 291

link 6, 31, 32, 47

output 178
ratio 126, 131
shrinkage
 complete 50, 52
 partial 50, 52

linkage 100

advantages 73
antiparallelogram 57
assemblability 64
basic building blocks 32, 63
cam-driven 798
circle-tracing 156
compliant 313
crank-rocker
 180 deg output 154
 360 deg output 154
deltoid 59
design 32
disadvantages 73
double-parallelogram 59

fourbar

independent parameters of 270

fourbar drag-link 153

Galloway 59

Grashof

inversions 57

Grashof condition 63

isocles 59

kite 59

large angular excursion 154

linkless 316

non-quick-return 119

parallelogram 57

rotatability 56

self-locking 103, 114. *See*

Also toggle

servo-driven 806

sixbar 207

Stephenson's 207

Watt's 207

special-case Grashof 57

substituted for gears 156

synthesis 100, 102, 134

torque 104, 111

transformation 43, 48, 409

linkages 6, 32

cascaded 153

connected in parallel 63

connected in series 63, 153, 154

versus cams 73

Linkages program

fivebar

coupler curves 131

exact straight line 144

fourbar

cognates 137

coupler curve 126

fivebar equivalent 141

quick-return 119

straight-line linkages 144

symmetrical linkage 131

three-position synthesis 111, 114

toggle 211

toggle positions 103

two-position synthesis 109

general information 817

linkage

force analysis 589

sixbar

double-dwell 153, 416

linkages 63

quick-return 122

single-dwell 151

linkless fourbar linkage 316

living hinge 66
 load
 lines 75
 torque 75
 load sharing 502
 locomotive 57
 Loerch 233, 257
 log roller 5
 Lord Kelvin
 comment on Peaucellier linkage 144
 losses 580, 772
 lower pair 6, 33. *See Also* joint
 lubricant 33
 lubrication 69, 73, 478, 677
 hydrodynamic 69
 problems 70
 seals for 69
 lumped
 mass 648
 model 690, 691
 model 571, 691, 763
 parameter 571, 575

M

machine 4, 5, 35, 178
 definition 4, 35
 design 3, 5, 73, 98, 385
 machinery
 rotating 559
 Maglev 797
 mandrel 662
 mass 4, 18, 554, 565, 687, 764, 774, 775
 balance 644
 density 555
 effective 573, 574, 775, 785
 equivalent 687
 lumped 574, 689
 moment 556, 652
 moment of inertia 558, 590, 618, 656, 687
 point 554, 562, 643
 masses
 combining 571
 massless rod 562, 643
 mass-radius product 645, 649, 663, 708
 mass-spring model 764

materials 710
 Mathcad 217, 464
 Matlab 464
 matrix
 augmented 247, 251, 259
 coefficient 251
 inverse 246
 solution 245, 577
 solver 245, 577
 Matrix program
 example
 force analysis 597, 599, 602
 linkage synthesis 255
 force analysis 577
 how to use.
 See programs: general information;
 See Also programs: how to run them
 solution method 247
 May, Rollo 21
 mechanical
 advantage 310, 493
 analog computer 100
 circuit 568
 efficiency 309
 engineering 5
 function generator 100
 system 569
 mechanism 4, 5, 35, 40, 178
 cam- vs. servo-driven 812
 compliant 65, 66
 advantages 66
 bistable 67
 crank-shaper 124
 definition 4, 35
 double-dwell 151
 forces in 553
 large angular excursion 154
 non-quick-return 119
 optical adjusting 312
 pick-and-place 154
 planar 101
 quick-return 119
 remote center 155
 washing machine 154
 Whitworth 124
 MEMS 67
 microchips 67
 microcomputer 14, 101
 Micro Electro-Mechanical Systems 67
 microgears 67

micromotor 67
 microsensors 68
 Milton, J. 490
 mirror symmetric 738, 752
 mks system 18
 mobility 37
 model 14, 252, 557, 568
 cardboard 14, 103, 107
 dynamic 562
 dynamically equivalent 688.
 See Also dynamic equivalence:
 requirements for
 finite-element 674
 lumped mass 691
 lumped parameter 764
 of rotating links 562
 simplified 554
 single DOF 576
 statically equivalent 690
 modeling
 rotating links 562
 modified
 sine 435
 trapezoid 427, 434
 module 499
 modulus
 of elasticity 574
 of rupture 574
 moment
 first of mass. *See* mass: moment
 first, of mass 556
 mass 556
 of inertia.
 See mass: moment of inertia
 definition 558
 experimental method 560
 transferring 559
 primary 736
 secondary 736, 739
 second of mass.
 See mass: moment of inertia
 second, of mass 558
 shaking 642, 672, 753
 cancelling 752
 in inline engines 728, 730
 in vee engines 744
 momentum 554, 677
 Monge, Gaspard 5
 motion
 complex
 definition 184
 generation 111, 234.

See Also function: generation;

See Also path: generation

analytical synthesis 235, 237, 247

definition 101

three-position 111, 112

two-position 107

intermittent 78

parallel 137

simple harmonic 421

straight-line 99

motor 38, 74, 582

AC 74

closed loop 78

compound-wound 74, 76

DC 74

permanent magnet (PM) 74

speed-controlled 77

garmotor 74, 78

micro 67

open loop 78

permanent magnet 74

series-wound 76

servo 74

shunt-wound 76

speed torque characteristic 617

stepper 74, 78, 101

synchronous 77

universal 74

movie camera 126

moving

centrode 313, 316.

See Also centrodes

pivots 112, 115, 151

multiple solutions 16

N

Nakamura 756

balancer 756

Nascar

engine redline 684

natural frequency 756, 766, 789

circular 766

damped 766, 768

undamped 766, 768

Nelson, G. L. 126

neural network 276

Newton-Raphson method 207, 214, 387

chaotic behavior of 215

in equation solvers 218

Newton's

equation 18, 764

laws 382, 553, 589, 643

method 214, 589

second law 3

third law 595

node 32, 38, 47

noise, vibration, and harshness (NVH) 756

no-load speeds 77

non-Grashof 56

triple-rocker 210

number synthesis 42

nut 33

O

objective function 270

octoid 508

offset 196

in slider-crank

definition 196

oil bath 478

open

kinematic chain 38

mechanism 193, 256. *See*

Also crossed: mechanism

operator 189

order

of joints 35

of links 42

of polynomial 439

orthogonal 196

oscillation - cam-follower 773

Otto cycle 675

overbalanced crank 708

overdamped. *See* damping

overdamped system 769

overlays 99

overshoot of response 769

oversquare engine 710

overtones 766, 772

overturning moment 465

P

pairs 33. *See Also* joint

higher 33

lower 33

pantograph 156

parabolic displacement 797

paradoxes

Ferguson's 526

Gruebler's 46

parallel

axis theorem 559, 624

connections

dampers 569

springs 569

linkage planes 101

motion 138

parallel motion 141

parallelogram linkage 57, 59

See Also antiparallelogram

particular solution 766, 771.

See Also homogeneous: solution

patent

crankshaft 141

websites 9

path 101

generation 101.

See Also function: generation;

See Also motion: generation

definition 101

precision points 236

with coupler

curve 125, 126, 128, 134

with prescribed timing 101, 266

pawl

driving 53

locking 53

Peaucellier 144

percussion. *See* center: of percussion

performance specifications 10, 15, 22, 639

phase angle

crankshaft 719

geared fivebar 131, 204

optimum 719

sign convention 722

physical pendulum 658

piecewise continuous function 422

piezoelectric

accelerometer 785

force transducer 662

pin

double shear 71

forces 684, 699

crankpin 700

mainpin 703

wristpin 700

joint 33, 69

single shear 71

pinion 492, 507

piston 384, 606, 674, 688, 691
 acceleration 680, 683, 693
 engine 55
 position 680
 pump 55, 606, 675, 717
 velocity 680

pitch
 circle 493, 495, 497
 curve 460, 467, 469
 diameters 493, 495
 diametral 498
 point 493, 494

pivots
 fixed-specified. *See* specified fixed
 pivots
 moving. *See* moving: pivots

planetary gear train.
See gear: train: epicyclic

planet gear 521. *See Also* sun gear

platform rocker 316

point masses. *See* mass: point

polar
 coordinates 188
 form 180
 plot 739. *See Also* hodograph

poles 298

polodes 313. *See Also* centrodes

polynomial 784
 345 439, 441, 442
 4567 442
 asymmetrical rise/fall 448
 3-segment 450
 function 420, 439
 design rule 446

POSE 183

position 178, 180, 291
 absolute 185
 analysis 185, 187
 difference 183
 equation 182, 293
 of any point on a link 207
 relative 183
 vector 180, 188

potential energy.
See energy: potential

pounds force (lbf) 18

pounds mass (lbm) 18

power 74, 309, 568, 580, 581,

731, 783
 equation 582
 stroke 677, 736, 738
 angles 731, 733
 to weight ratio 675

practical considerations 624, 789, 812

precision
 points 236, 243
 position 236, 248

preload
 cam-follower spring 775.
See Also spring: preload

preloaded structure 40

pressure angle
 cam-follower 460
 flat-faced 465
 force analysis 785
 roller 461
 of gearsets 495

primary component.
See Also Fourier: series
 of shaking force 683
 of shaking moment 730

prime
 circle 460, 785
 radius 460, 468

principal axes 31

principle
 of d'Alembert 578
 of transmissibility 307

problem
 definition 22
 unstructured 8

production 14

programs 817
 disclaimer on liability 818
 Dynacam 817, 818.
See Also Dynacam program
 Engine. *See* Engine program
 Linkages 817, 818
 Matrix 817.
See Also Matrix program

Projects
 Chapter 3 173
 Chapter 8 485
 Chapter 11 639
 Chapter 13 716
 Chapter 14 761

prototypes 13, 575

prototyping 13
 and testing 13

publications, technical
 websites for 9

pulleys 509

pulse. *See* jerk

pure
 harmonic 621, 730, 767
 rolling 35, 125
 joint 35
 rotation 32, 107, 291, 688, 691
 slide 35
 translation 32, 370, 563, 688, 691

pushrod 574

Q

qualitative
 synthesis. *See* synthesis: qualitative

quasi-static 578

quaternary link 32, 42

quick
 forward 120
 return 119, 122, 126
 mechanism 55
 sixbar 121

R

rack 507
 and pinion 507
 steering 507

radius
 of curvature 460
 flat follower 470
 roller follower 466, 467
 of gyration 561, 565
 prime circle 777

ratchet 6
 and pawl 53
 wheel 53

ratio
 gear 571
 lever 571

reference frame 36.
See Also coordinate system

relative position
See position: relative

report, technical 17

resistor 569. *See Also* damper

resonance 772, 796
 cam-follower

- force-closed* 774
- form-closed* 780
- resonate** 772
- response**
 - complete 772
 - damped 767
 - forced 772
 - steady state 772
 - transient 766, 769, 772
 - undamped 764
- Reuleaux, Franz** 6, 33, 38
 - classification of mechanisms 35
- reverted**
 - compound train 514.
 - See Also* gear: train
 - gear train design 515
- revolvability** 64
 - definition 64
- right angle drives** 507.
 - See Also* gear: bevel
- rigid body** 31, 32
 - acceleration 364
 - motion 183
- ring gear** 521.
 - See Also* gear: train: epicyclic
- rise-dwell-fall-dwell.** *See* cam
- rise-dwell-fall-dwell cam** 410, 415
- rise-fall cam** 410, 415
- rise-fall-dwell cam** 410, 415, 443
- Roberts**
 - Chebyshev theorem 134.
 - See Also* Chebyshev
 - diagram 134.
 - See Also* Cayley diagram;
 - See Also* cognate
 - straight-line linkage 141
- Roberts, Richard** 141
- Roberts, Samuel** 134, 141
- robot** 38, 100
- rocker**
 - arm 574
 - definition 36
 - infinitely long 52
 - output 105
- rocking**
 - chair 316. *See Also* centrodes
 - couple 646
- roller** 413.
 - chain 510
 - follower 413, 468, 476, 565, 764, 775
- chrome plated* 477
- crowned* 477
- in valve trains* 477
- materials* 477
- slip* 476
- rolling**
 - centrodes 508
 - cones 507
 - contact 6
 - cylinders 491, 493
- roll-slide joint** 33, 35
- root finding** 387
- rotatability** 56, 63, 64
 - definition 63
 - of geared fivebar linkage 63
 - of N-bar linkages 64
- rotation** 31
 - definition 184
 - pure 32
 - balance in 642
- rotational**
 - DOF 33
 - freedom 35
 - kinetic energy 675
- rotopole** 107
- roughness.** *See* surface contact

S

- Sanders** 387
- Sandor, G. N.** 6, 119, 233
- scalar magnitude** 306
- scaling** 13
- SCCA**
 - family of curves 429, 433
- Scotch yoke** 52, 53
- screw** 6
 - joint 33
- s-curve** 797
- second**
 - harmonic 683
 - moment
 - of area 574
 - of mass 558
- secondary component.**
 - See Also* Fourier: series
 - of shaking force 683
 - of shaking moment 730
- selection** 13
- self-locking linkage.**
 - See* linkage: self-locking
- series connections**
 - dampers 569
 - springs 570
- servo**
 - mechanism 497, 795
 - motor 78, 101, 795, 796
 - linear 796
 - valve 79
- shaft**
 - encoder 662
 - hollow 521
- shaking**
 - force. *See* force: shaking
 - moment. *See* moment: shaking
 - torque. *See* torque: shaking
- sheave** 509. *See Also* pulleys
- shock** 385. *See Also* jerk
- silent chain.** *See* chain
- simple**
 - gear train. *See* gear: train: simple
- simply supported.**
 - See* beam: simply supported
- simultaneous equation solution** 245.
 - See Also* Matrix
 - dynamic forces 577, 590
- single**
 - cylinder engine 674
 - dwell 153, 443
 - cam 443
 - mechanism 148, 149, 416
 - enveloping wormset 506
 - gearset 511
- single-plane balance.**
 - See* balance: single-plane
- SI system** 17
- sixbar**
 - drag-link quick-return 121, 194, 197, 200, 323, 368
 - linkage 134
 - mechanism 148
 - Watts linkage 109
- Sixbar program**
 - example
 - three-position synthesis 257
- skew axis** 31
- slide**
 - ball-bearing 73
 - linear 73
- slider block** 50, 606
- slider-crank** 50, 55, 74, 122, 301, 606, 608

analysis
 acceleration 369
 Fourier 683
 instant centers 301
 position solution 196
 vector loop 196
 inverted 201, 376, 608
 acceleration 376
 Whitworth crank-shaper 55
 linkage
 dynamic model 691
 in IC engines 672
 multicylinder 717
 offset 201
 one-cylinder 674
 nonoffset 679
 offset
 definition 196
sliding
 contact 6
 joints 43. *See Also* joint
slip 125
 component 319, 320, 374
 velocity 373
slop 497
slugs 17, 18
solenoid 38, 79, 100
solids modelling 624
solution methods 577
Soni, A. 138
source torque. *See* torque: source
space probe lost 18
space width (gears) 497, 498
spatial
 linkage 128
 mechanisms 101
specified fixed pivots 114, 117, 257.
 See Also pivots: fixed: specified;
 See Also synthesis: of linkages
speed controlled DC motor.
 See motor: DC
speed-torque characteristic.
 See motor
spline functions 460
spreadsheet 577
spring
 compression 574
 constant 65, 565, 574, 764,
 775, 780
 definition 566
 effective 570, 573, 775
 free length 775

 helical coil 776
 physical 775, 780
 preload 777
springs 129, 570
 as links 65
 combining 570
 in parallel 571
 in series 570
sprocket 510
standard form equation 266, 268.
 See Also analytical linkage
 synthesis
static
 balancing. *See* balancing: static
 equivalence 690
 friction 566
steady state 766
Stephenson's
 sixbar 207
stiction 566
stiffness 555, 774
stops 53, 435. *See Also* dwell
straight-line
 linkage 5, 99
 Chebyshev 141
 Evans 144
 exact 144
 Hart 144
 Hoeken 143
 optimum 144
 Peaucellier 144
 Roberts' 141
 Watt' 141
 mechanisms 141
 approximate 144
 exact 144
strain gages 785
strength 555
stresses 4, 178, 384, 555
structural
 building block 46
 subchain 48
structure 40, 42
 preloaded 42
Suh, N. P. 15
sun gear 521.
 See Also gear: train: epicyclic
superposition 577, 578, 685, 699
surface contact 33
suspension system 311
s v a j diagrams 416
 polynomials 439

sweet spot 563.
 See Also center: of percussion
synchromesh 506, 532
 clutch 532
 transmission 532
synchronous belt.
 See belt: synchronous
synonyms 11
synthesis 98, 99, 102, 106
 algorithm 100
 analytical 100, 104
 compare to graphical 243
 elastic energy method 275
 equation 270
 equation methods 273
 optimization methods 273
 optimized 270
 precision 270
 precision point methods 272
 selective precision synthesis 274
 using genetic algorithms 275
 definition 8
 four-position
 analytical 266
 graphical 119
 graphical
 tools needed for 105
 of mechanisms 3, 25, 30
 qualitative 98, 99, 100, 106
 quantitative 100, 119
 three position
 analytical 247, 251, 253
 graphical 111
 motion 248
 specified fixed pivots 257, 260
 two position 105
 analytical 243
 graphical 111
 type 100

T

tabular method 523
tackle 6
tappet 574
TDC. *See* top dead center (TDC)
ternary link 32, 42, 44.
 See Also link
testing 13, 14
thermodynamics 672, 679
threebar crank-slider 593.
 See Also slider-crank

three-freedom joint 33.
See Also joint

three-position synthesis.
See synthesis: three position: motion

through variable 568

time ratio 119, 121, 122

timing
 belt. *See* belt: timing
 diagram 417

Ting, K. L. 63

TKSolver 217, 387, 464, 624

toggle 102, 103, 314
 angle 212
 linkage 103
 position 102, 114, 134, 236, 243
 calculating location 210
 in rock crusher 311

tolerance
 (human) of acceleration 383

toothed belts 509. *See Also* belt

tooth thickness 497, 498

top dead center (TDC) 675

torque 656
 applied 591
 camshaft 783
 converter 533
 lock-up clutch 534
 stator blades 534
 driving 581
 dynamic 783, 785
 external 581
 flywheel-smoothed 697
 gas 686
 flywheel-smoothed 697
 in four-cylinder inline engine 733, 739
 in one-cylinder engine 685
 in vee engines 746
 inertia 685
 flywheel-smoothed 697
 in four-cylinder inline engine 736, 739
 in inline engines 727
 in one-cylinder engine 695
 in vee engines 745
 in virtual work 581
 input 657, 783
 oscillations in 783
 ratio 310, 493
 shaking 612, 685
 in fourbar linkage 657
 in multicylinder engines 730

in one-cylinder engines 694
 source 591, 596, 601
 total 739, 750
 engine 697
 variation 616

torque-speed relation 74. *See*
Also speed-torque characteristic

torque-time
 diagram 619
 function 697

Towfigh, K. 21, 22, 312

trade-offs 100, 148, 442, 780

train ratio 512. *See Also* gear: ratio

transducer 795

transfer
 port 677
 theorem 559

transient 766

translating
 follower 410
 slider 121

translation 31, 32
 curvilinear 137, 183
 definition 183
 rectilinear 183, 411

translational DOF 33

transmission 493, 506, 675
 automatic 526
 automotive 532
 component 319, 320, 374
 compound epicyclic manual 535
 continuously variable 535
 Ford model T 535
 synchromesh 532

transmission angle 307, 309, 310, 568, 622
 definition 103, 208
 differences in cognates 137
 extreme values 209
 limited application 208
 minimum 104
 optimal 122
 poor 243
 quick-return linkage 121

trapezoidal
 acceleration 426
 rule 620

tricircular sextic 124, 273

triple-rocker 57, 210

truss 48

two-bar chain 105, 121

two-dimensional space 31

two-freedom joint 33

two-plane balance 646

two position synthesis.
See synthesis: two position

two-stroke
 cycle 675, 677, 731
 engine 732

type synthesis 99, 158.
See Also synthesis: type

U

undercutting 467, 470, 500

underdamped 770, 771.
See Also damping

undersquare engine 710

units 17
 space probe lost due to 18

units systems 17

unit vectors 188

Unobtainium 688, 774

unstructured problem 8

V

valve 574
 cam 476, 492
 float 776
 spring 574

valves 409

vector
 angle of
 definition 189.
 See Also free vector
 loop 238

vee
 angle 721, 739, 742, 750. *See*
 Also bank: angle;
 See Also engines: vee
 desirable 750
 belt 509.
 See Also belt: vee
 engines.
 See engines: vee

velocity 178, 291
 absolute 292, 297, 298
 analysis
 algebraic 321
 geared fivebar 330

graphical 294, 296
inverted slider-crank 319
sliding joint 318
using instant centers 305
 angular 291, 319, 330, 375
 cam 416
 cam-follower
 peak factor 432
 constant 143
 in slider-crank 153
 definition 291
 difference 293, 297, 298, 321, 359
 equation 293
 of point on link 331
 of slip 317, 330, 375
 of transmission 330
 ratio 493, 499, 508
 of involute gears 497
 relative 293, 297, 317, 321, 566
vibration
 in cam-followers 568, 771, 780
 in engines 730
 in linkages 612
videos 26
virtual laboratory 26, 408, 485, 638, 671
virtual work 582, 613
 equation 582
ViseGrip 311
visualization 24
voltage 568

W

Wampler, C. 203, 272
Watt, James 5, 141
 proudest accomplishment 141
Watt's
 epicyclic crank 141
 linkage
 guide steam engine 141
 sixbar 105, 207
 straight-line fourbar 141
wear 33
weather systems 375
wedge 5
weight 18
weighting factor 13
well pump 55
wheel 6
 and axle 5

Whitworth quick-return 55, 124
Willis, Robert 5
windshield wiper linkage 59
Wood, George A. Jr. 21
work 36, 580
Working Model 99, 103, 817
worldwide web 9
 keywords for searching 29
 useful sites 29
worm 506
 lead angle 538
 set 506
 wheel 506
wrapping connectors 6
wrist pin 689, 691
writing engineering reports 29

Y

Young's modulus 574

Z

zero velocity 126
Zhang, C. 131

DOWNLOADS INDEX

ANIMATIONS Folder

AVI, Working Model, and Matlab files by Sid Wang

These files are self-cataloging. Run the master catalog file Animation.html to access and run these animations. Most have AVI movie files in addition to their native file formats. The native Working Model files also can be accessed directly from the Working Model Files folder listed below.

CUSTOM PROGRAMS

Programs by R. L. Norton

PROGRAM DYNACAM

PROGRAM LINKAGES

PROGRAM MATRIX

These are available from the author's website at www.designofmachinery.com. Print-book users can register on my website as a student or professor, and I will send a password to access a protected site where they can download the latest versions of these programs. Student or professor registration will also allow print-book users to download all the files listed in this index. Digital book users will have access to the downloadable files in the Video Contents and in this Index, and the computer programs through the publisher's website.

Note that I personally review each of these requests for access to my protected site and will approve only those that are filled out completely and correctly according to the provided instructions. I require complete information and accept ONLY university email addresses for both you and your instructor. (No Gmail, Yahoo, Naver, etc.) So be sure to follow the instructions exactly, or your request will be denied.

Run the Install.exe file to install the program.

EXAMPLES AND FIGURES Folder

Data files for Norton's custom programs that match some examples and figures in text.

Chapter 2 Subfolder

F02-19b.5br

Chapter 3 Subfolder

Cognate1.4br
Cognate2.4br
Cognate3.4br
F03-01a.4br
F03-01b.4br
F03-04.4br
F03-06.4br
F03-07b.6br
F03-07c.6br
F03-08.4br
F03-09c.6br
F03-12.4br
F03-13a.6br
F03-17b.4br
F03-18.4br
F03-24.4br
F03-28a.4br
F03-28b.5br
F03-29a.4br
F03-29c.4br
F03-29d.4br
F03-29e.4br
F03-29f.4br
F03-31c.6br
F03-34.6br
F03-35.6br
FP03-07.4br
Straight.5br

Chapter 4 Subfolder

F04-11.5br
F04-15.4br

Chapter 5 Subfolder

E05-01.4br
E05-02a.mtr
E05-02b.mtr
E05-02.4br
E05-03.4br

Chapter 6 Subfolder

F06-14.4br
F06-15a.4br

KEY TO FILENAME SUFFIXES

DYNACAM	.CAM
ENGINE*	.ENG
FIVEBAR*	.5BR
FOURBAR*	.4BR
LINKAGES	.BAR
MATLAB	.M
MATRIX	.MTX
SIXBAR*	.6BR
SLIDER*	.SLD
TKSOLVER	.TKW
WORKING MODEL	.WM2D, WM3

* Program LINKAGES will open these files.

F06-15b.4br
F06-17b.4br

Chapter 8 Subfolder

E08-03.cam
E08-04.cam
E08-05.cam
E08-06.cam
E08-07.cam
E08-08.cam
E08-09a.cam
E08-09b.cam
E08-10a.cam
E08-10b.cam
E08-10c.cam
E08-11.cam
E08-12.cam

Chapter 11 Subfolder

E11-01.mtr
E11-02.mtr
E11-03.mtr
E11-03.4br
F11-06.4br

Chapter 12 Subfolder

F12-05.4br

Chapter 14 Subfolder

BMWV12.eng
F14-12.eng
F14-14.eng
F14-18.eng
F14-24.eng

Chapter 15 Subfolder

E15-01.cam
E15-02.cam

Appendix A Subfolder

F_A-05.4br
F_A-11.5br

LINKAGE ATLASES Folder

Contains PDF file of atlases of coupler curves for fourbar and geared fivebar linkages.

Hrones and Nelson Fourbar Atlas

Zhang et al. Geared Fivebar Atlas

PDF PROBLEM WORKBOOK Folder

Contains PDF files of all the figures needed to solve the text's end-of-chapter problems. Each PDF file contains one problem figure and all of the problem statements associated with it. They are grouped in subfolders by chapter and their filenames are the same as the figure number or problem number involved. These files provide the student with a printable workbook of illustrated problems in which graphical problem solutions can be directly worked out or analytical solution results recorded.

PROBLEM SOLUTIONS Folder

Data files that solve selected problems in the text.

Chapter 3 Subfolder

P03-14.4br
P03-22.4br
P03-23.4br
P03-36.4br
P03-42.4br

Chapter 4 Subfolder

P04-21.4br
P04-23.4br
P04-25.4br
P04-26.4br
P04-29.4br
P04-30.4br

Chapter 5 Subfolder

P05-08.4br
P05-11.4br
P05-15.4br
P05-19.4br
P05-26.4br

Chapter 6 Subfolder

P06-47.4br
P06-48.4br
P06-49.4br
P06-51.4br
P06-62.4br

Chapter 7 Subfolder

P07-04a.4br
P07-04c.4br
P07-04e.4br

P07-04g.4br
P07-04i.4br
P07-04k.4br
P07-04m.4br
P07-39.4br
P07-40.4br
P07-41.4br
P07-42.4br
P07-44.4br

Chapter 10 Subfolder

P10-04a.mtr
P10-04b.mtr

Chapter 11 Subfolder

P11-03a.sld
P11-03c.sld
P11-03e.sld
P11-03g.sld
P11-04a.tkw
P11-05a.tkw
P11-05a.4br
P11-05c.4br
P11-05e.4br
P11-05g.4br
P11-06a.tkw
P11-06c.tkw
P11-06e.tkw
P11-06g.tkw
P11-07a.4br
P11-07c.4br
P11-07e.4br
P11-12.4br
P11-13.4br

Chapter 12 Subfolder

P12-09.4br

Chapter 13 Subfolder

P13-11.eng
P13-14.eng
P13-19a.eng
P13-19b.eng

PROGRAM MANUAL Folder

Contains a PDF file of the user manual for programs LINKAGES, DYNACAM, and MATRIX.

TKSOLVER FILES Folder**TKSolver model files.**

The TKSolver program is needed to run these files and is not included with this text. See www.uts.com.

Gears.tk Subfolder

Compound.tkw
Revert.tkw
Triple.tkw

Linkages.tk Subfolder

3 position FixPivots.tkw

3 position.tkw
Cognate.tkw
Coupler.tkw
DragSlider.tkw
Eq04-02.tkw
Ex11-04.tkw
Figure P05-05.tkw
Fivebar.tkw
Fourbar.tkw
Inverted slider-crank.tkw
SCCA.tkw
Slider_Cmpr.tkw
Slider.tkw
Soni Cognate.tkw
Symmetric.tkw
Transport.tkw
Virtual Work.tkw

Misc.tk Subfolder

CamCalc.tkw
Constrnt.tkw
Cubic.tkw
Cycloid.tkw
F04-18.tkw
Pressang.tkw
SCCA.tkw
Student.tkw

VIDEOS

See the Video Contents.

VIRTUAL LABS

See the Video Contents.

WORKING MODEL FILES Folder**Chapter 2** SubfolderWorking Model 2D Files

02-10b.wm2d - Scotch Yoke
02-12a.wm2d - Geneva
02-12b.wm2d - Ratchet and Pawl
02-12c.wm2d - Linear Geneva
02-13.wm2d - Slider-Crank
02-14abc.wm2d - Stephenson Inversion
02-14de.wm2d - Watt Inversions
02-15.wm2d - Grashof Inversions
02-16.wm2d - Non-Grashof Inversions
02-19b.wm2d - Geared Fivebar
02-20.wm2d - Desk Lamp
P2-01f.wm2d - Overhead Valve
P2-03.wm2d - Front End Loader
P2-04c.wm2d - Radial Engine
P2-04d.wm2d - Walking Beam
P2-04e.wm2d - Drafting Arm
P2-04g.wm2d - Drum Brake
P2-04h.wm2d - Compression Chamber
P2-05a.wm2d - Chebyshev Mechanism
P2-05b.wm2d - Kempe SL Mechanism
P2-07.wm2d - Throttle Mechanism

P2-08.wm2d - Scissors Jack
 P2-10.wm2d - Watt's Engine
 P2-13.wm2d - Crimping Tool
 P2-14.wm2d - Pick and Place
 P2-15.wm2d - Power Hacksaw
 P2-16.wm2d - Powder Press
 P2-18.wm2d - Oil Field Pump

Working Model 3D Files

P2-01h.wm3 - Cylindrical Cam

Chapter 3 Subfolder

Working Model 2D Files

03-04.wm2d - Example 3-1
 03-05.wm2d - Example 3-2
 03-07b.wm2d - Example 3-4
 03-09c.wm2d - Example 3-6
 03-11.wm2d - 3-Position Synthesis
 03-12b.wm2d - 4br Quick Return
 03-13a.wm2d - 6br Quick Return
 03-14.wm2d - Quick-Return Shaper
 03-14.*.wm2d - Quick-Return Shaper
 03-15.wm2d - Coupler Curves
 03-17.wm2d - Coupler Curve Atlas
 03-17a.wm2d - Coupler Curve Atlas
 03-18.wm2d - Camera Film Advance
 03-18.*.wm2d - Camera Film Advance
 03-19a.wm2d - Auto Suspensions
 03-19a.*.wm2d - Auto Suspensions
 03-24a.wm2d - Roberts Diagram
 03-25a.wm2d - Roberts Diagram
 03-25b.wm2d - Roberts Diagram
 03-26.wm2d - Chebyshev Cognates
 03-26a.wm2d - Roberts Diagram
 03-26b.wm2d - Chebyshev Cognates
 03-26b.*.wm2d - Chebyshev Cognates
 03-27c.wm2d - Curvilinear Trans.
 03-27d.wm2d - Curvilinear Trans.
 03-28.wm2d - GFBM 4br Cognate
 03-28.*.wm2d - GFBM Cognates (alt.)
 03-29.wm2d - Straight-Line Linkages
 03-29a.wm2d - Watt Straight-Line
 03-29b.wm2d - Watt's Engine
 03-29c.wm2d - Roberts Straight-Line
 03-29d.wm2d - Chebyshev SL
 03-29e.wm2d - Hoeken Straight-Line
 03-29f.wm2d - Evans Straight-Line
 03-29g.wm2d - Peaucellier Strt-Line
 03-31c.wm2d - Single-Dwell—Rocker
 03-31d.wm2d - Single-Dwell—Slider
 03-32.wm2d - Double-Dwell Linkage
 03-34.wm2d - 180° Rocker Output
 03-35.wm2d - Washing Machine
 03-36.wm2d - 360° Rocker Output
 P3-03.wm2d - Treadle Wheel
 P3-07.wm2d - Walking Beam
 P3-08.wm2d - Loom Laybar Drive

Chapter 4 Subfolder

Working Model 2D Files

04-16.wm2d - Double Rocker Toggle
 P4-01.wm2d - Fourbar Analysis
 P4-02.wm2d - Slider-Crank Analysis
 P4-03.wm2d - Inverted Slider-Crank
 P4-05c.wm2d - Radial Engine
 P4-05d.wm2d - Walking Beam
 P4-05e.wm2d - Drafting Machine
 P4-05g.wm2d - Drum Brake
 P4-05h.wm2d - Compression Chamber
 P4-06.wm2d - Pick and Place
 P4-07.wm2d - Power Hacksaw
 P4-09.wm2d - Walking Beam Conveyor
 P4-11.wm2d - Loom Laybar Drive
 P4-14.wm2d - Treadle Wheel
 P4-18.wm2d - Elliptical Trammel

Chapter 6 Subfolder

Working Model 2D Files

06-05c.wm2d - Instant Centers
 06-10b.wm2d - Instant Centers
 06-11.wm2d - Rock Crusher
 06-12.wm2d - Suspension
 06-14a.wm2d - Centrodes 1
 06-14b.wm2d - Centrodes 2
 06-14c.wm2d - Centrodes 3
 06-14d.wm2d - Centrodes 4
 06-15a.wm2d - Centrodes 5
 06-15b.wm2d - Centrodes 6
 06-17a.wm2d - Cycloidal Motion
 P6-01.wm2d - Fourbar Analysis
 P6-02.wm2d - Slider-Crank Analysis
 P6-03.wm2d - Inverted Slider-Crank
 P6-08c.wm2d - Radial Engine
 P6-08d.wm2d - Walking Beam
 P6-08e.wm2d - Drafting Machine
 P6-08g.wm2d - Drum Brake
 P6-08h.wm2d - Compression Chamber
 P6-15.wm2d - Power Hacksaw
 P6-16.wm2d - Pick and Place
 P6-18.wm2d - Powder Press
 P6-19.wm2d - Walking Beam Conveyor
 P6-21.wm2d - Toggle Pliers
 P6-23.wm2d - Surface Grinder
 P6-29.wm2d - Drum Pedal
 P6-30.wm2d - Oil Field Pump
 P6-32.wm2d - Elliptical Trammel

Working Model 3D Files

06-12.wm3 - Bump Steering

Chapter 7 Subfolder

Working Model 2D Files

P7-01.wm2d - Fourbar Analysis
 P7-02.wm2d - Slider-Crank Analysis
 P7-03.wm2d - Inverted Slider-Crank
 P7-08c.wm2d - Radial Engine

P7-08d.wm2d - Walking Beam
 P7-08e.wm2d - Drafting Machine
 P7-08g.wm2d - Drum Brake
 P7-08h.wm2d - Compress Chamber
 P7-15.wm2d - Power Hacksaw
 P7-16.wm2d - Pick and Place
 P7-19.wm2d - Walking Beam
 P7-20.wm2d - Surface Grinder
 P7-24.wm2d - Drum Pedal

Chapter 8 Subfolder

Working Model 2D Files

08-02a.wm2d - Translating Follower
 08-02b.wm2d - Oscillating Follower
 08-03a.wm2d - Roller Follower
 08-03c.wm2d - Flat-Faced Follower
 08-39.wm2d - Cam and Follower
 08-48.wm2d - Radii of Curvature
 E8-02.wm2d - Example 8-2
 E8-03.wm2d - Example 8-3
 E8-04.wm2d - Example 8-4
 E8-07.wm2d - Example 8-7

Working Model 3D Files

08-03a.wm3 - Roller Follower
 08-04.wm3 - Cylindrical Cam

Chapter 9 Subfolder

Working Model 2D Files

09-01b.wm2d - Internal Gearset
 09-04.wm2d - External Gearset
 09-05.wm2d - Involute Curves
 09-06.wm2d - Tooth Engagement
 09-19.wm2d - Rack and Pinion
 09-28.wm2d - Compound Gear Train
 09-33.wm2d - Planetary Gearset

Working Model 3D Files

09-16.wm3 - Helical-Parallel Gears
 09-17.wm3 - Helical-Crossed Gears
 09-18.wm3 - Worm and Worm Gear
 09-21.wm3 - Bevel Gears
 09-30.wm3 - Gear Trains
 09-34.wm3 - Planetary Gearset
 09-44a.wm3 - Transmission - High
 09-44b.wm3 - Transmission - Low
 09-44c.wm3 - Transmission - Reverse
 09-51.wm3 - Drive Train
 P9-02.wm3 - Compound Epicyclic
 P9-03_open.wm3 - Differential
 P9-03_locked.wm3 - Differential

Chapter 10 Subfolder

Working Model 2D Files

10-11a.wm2d - Valve Train

Chapter 13 Subfolder

Working Model 2D Files

13-01.wm2d - Vee-Eight Engine

